

Dynamic multitasking and managerial investment incentives[★]

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Abstract

We study non-contractible intangible investment in a dynamic agency model with multitasking. The manager's short-term task determines current performance, which deteriorates with investment in the firm's future profitability, his long-term task. The optimal contract dynamically balances incentives for short- and long-term performance. Investment is distorted upwards (downwards) relative to first-best in firms with high (low) returns to investment. These distortions decrease as good performance relaxes endogenous financial constraints, implying negative (positive) investment-cash flow sensitivities. Our results shed light on how corporate investment policies, liquidity management, and executive compensation structure differ across industries with different returns to intangible investment.

Keywords: Continuous time contracting, Multiple tasks, Delegated investment, Managerial compensation, Investment dynamics
JEL: D86, D92, J33

1. Introduction

A manager responsible for a firm's operations usually has some form of discretion in running the day-to-day business, which relies on his specific skills or private information. Due to the separation of ownership and control, an agency problem arises, which has been the focus of much of the recent dynamic financial contracting literature (e.g., [DeMarzo and Sannikov, 2006](#); [Biais et al., 2007](#)). Yet, in a dynamic world, firms also have to make strategic decisions and invest to maintain long-term profitability. Typically, this investment process also relies on information and skills of the same manager or is even delegated to him with discretion. In fact, many strategic investments, in particular those in intangibles such as investment in R&D, process innovation, product development, human capital, or distribution systems, share the following features: i) actual

investment expenditures are not easily verifiable by firm owners as they are either not reported or reported expenses can be manipulated;³ ii) their outcome is uncertain; and iii) they have persistent effects on the firm's profitability. The manager's hidden actions, thus, affect both the firm's current period payoffs as well as its long-term profitability, giving rise to a dynamic multitask problem.

To capture these ideas, we introduce delegated non-contractible investment in the firm's future profitability — which we interpret as investment in intangible capital — into a continuous time cash flow diversion model. We analyze how the need to incentivize the manager to meet both short-term and long-term targets affects the optimal compensation scheme, as well as the efficiency of the investment process. Under the optimal long-term contract, investment depends on the entire history of past performance and is distorted away from the first-best level. We show that both the sign of the investment distortion as well as the comparative statics of investment (with respect to realized cash flows, corporate governance or financial frictions) crucially depend on the technological returns to investment (i.e., the extent to which a firm's long-term profitabil-

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³Indeed, several forms of intangible or "soft" investment expenditures are — unlike hard investments in plants, property or other equipment — not reported explicitly in firms' financial statements. For those intangible investments that are reported, accounting rules often leave considerable discretion to managers "[...] to alter financial reports to either mislead some stakeholders about the underlying economic performance of the company, or to influence contractual outcomes that depend on reported accounting numbers" ([Healy and Whalen, 1999](#)), a practice referred to as earnings management. For instance, while R&D expenditures are reported in firms' income statements, "[...] the notion of what outlays are considered R&D [...] can be difficult to assess, and often represents the manager's discretionary choice" ([Koh and Reeb, 2015](#)). Exploiting this discretion, for example by shifting core expenses to special items such as R&D, is referred to as classification shifting in the accounting literature (see, e.g., [McVay, 2006](#); [Skaife et al., 2013](#); [Darrrough et al., 2017](#)).

ity depends on successful investment). Investment is distorted upwards relative to the first-best benchmark when returns to investment are high and it is distorted downwards when they are low. These distortions decrease in absolute terms with financial slack and, thus, past performance. In line with recent empirical evidence (Hovakimian, 2009; Peters and Taylor, 2017), this implies a positive relation between intangible investment and realized cash flows in industries with low, and a negative relation in industries with high returns to such investment. Further, if the costs of raising external funds increase, so will both the distortions in investment as well as the sensitivity of investment to past performance. Our multitask model with non-contractible investment, thus provides a new perspective on investment-cash flow sensitivities as a measure of financial constraints accounting for investment in intangibles (see, e.g., Brown and Petersen, 2009; Chen and Chen, 2012, for empirical evidence).

The key agency frictions in our dynamic multitask theory of investment are due to i) unobservable cash flows and ii) non-contractibility of actual investment expenditures. An agency problem with respect to investment then arises endogenously as a result of compensation for short-term performance, which is necessary to deter the manager from diverting cash flows for private consumption. Since his compensation is tied to current earnings figures, the manager has an incentive to cut profitable investment to boost the firm's short-term profits at the expense of its long-term profitability. Our model, thus, shares with the literature on managerial short-sightedness (e.g., Narayanan, 1985; Stein, 1988, 1989; von Thadden, 1995) the notion that short-term incentive schemes can induce managers to focus excessively on current period outcomes. However, in our model such unintended consequences of short-term incentive pay can be mitigated by also tying compensation to indicators of investment success, such as project milestones or the number of patents granted as is common practice in particular in high-tech firms (e.g., Balkin et al., 2000).

Yet, as is well known from related (single-task) dynamic financial contracting models (e.g., DeMarzo and Sannikov, 2006), exposing the manager to compensation risk to provide incentives is costly. In our model with bilateral risk neutrality, this formally is a consequence of limited liability, which requires termination of the contract if, after a series of bad outcomes, the manager is "too poor to be punished." In this case, he must be replaced, which is costly ex-post, but part of the ex-ante optimal long-term contract. Hence, the need to tie the manager's pay to (imperfect) signals of investment success or failure, such as to incentivize investment, creates additional agency costs. The optimal investment schedule is then determined by trading-off the agency costs of investment with the potential efficiency gains as captured by the respective returns to investment.

We show that the agency costs of investment are non-monotonic in the implemented investment level. In particular, marginal agency costs are positive for low and negative for high levels of investment. As a consequence, investment is distorted downwards relative to the first-best (owner-manager) benchmark in firms with low returns to investment where first-best investment is low. By contrast, investment is distorted upwards

relative to first-best when returns to investment and, thus, also first-best investment is high. To understand this result, consider a firm with high returns to investment and assume that the principal wants to implement the first-best investment level. Since the probability of an investment success is high given the high first-best investment level, and the probability of failure correspondingly low, a reward for success provides less incentives per unit of expected pay than punishment upon failure which is captured by a lower likelihood ratio. Accordingly, the optimal compensation policy relies heavily on punishment, such that the costs of providing incentives are concentrated on the failure event. Thus, if returns to investment are sufficiently high, agency costs can be reduced by distorting investment above the high first-best level, as this reduces the probability of having to bear the high incentive costs associated with punishing the manager for an investment failure. Similarly, if returns to investment are low, incentives are mainly provided by generous rewards. In this case, agency costs can be reduced by distorting investment downwards relative to the low first-best level, such as to reduce the probability of triggering a costly reward following investment success.

While the sign of investment distortions in our model is determined by the profitability of the investment technology, their size depends on the firm's past performance, creating novel implications for investment dynamics. In particular, when the manager's track record — the history of cash flows and investment outcomes under his tenure — improves, his stake in the firm under the optimal contract increases, mitigating the agency problem and, thus, reducing investment distortions. Accordingly, investment expenditures are negatively related to cash flows in overinvesting firms (i.e., in firms with high returns to investment), while in underinvesting firms (i.e., those with low returns to investment), the relation is positive.

We show that this dependence of investment on past performance is stronger the more severe the firm's financial constraints. Concretely, we consider a standard implementation of the optimal contract in which the manager's track record is captured by a measure of the firm's financial slack in the form of cash holdings (see, e.g., Biais et al., 2007). When cash holdings are depleted, the firm is no longer able to honor its payments to investors, triggering a costly restructuring of the firm. This involves replacing the incumbent manager and raising new external funds. We show that if the costs of raising external funds (i.e., financial constraints) increase, investment distortions become more severe and investment becomes more sensitive to cash flows for any given level of financial slack. Hence, investment increases and its sensitivity to financial slack becomes more negative in firms with high returns to investment, while investment decreases and its sensitivity to financial slack becomes more positive in firms with low returns to investment.

Our model, thus contributes to the debate about the relationship between investment-cash flow sensitivities and financial constraints, which here arise endogenously from the agency problem. By focusing on investment in future profitability per unit of (physical) capital instead of standard capital investment, our model addresses increasing importance of intangible investment, for example in R&D or knowledge and organiza-

tional capital (e.g., [Brown and Petersen, 2009](#)). We show that investment-cash flow sensitivities as an empirical measure of financial constraints need to be interpreted with caution to account for the distinct features of intangible investment. There are two main reasons. First, financially constrained firms may feature both positive and negative investment-cash flow sensitivities depending on firm- or industry-specific returns to intangible investment. Concretely, our model predicts investment-cash flow sensitivities to be negative in industries where intangible investment is particularly important (see also recent empirical evidence in [Hovakimian, 2009](#); [Peters and Taylor, 2017](#)) and positive else. Hence, our model offers an explanation for the puzzling findings in [Chen and Chen \(2012\)](#) that on average (i.e., not controlling for intangible investment intensity), investment-cash flow sensitivities have disappeared despite financial constraints still being a first-order concern for firms.

The second empirical challenge is that, in light of our model of non-contractible (intangible) investment, accounting measures of intangible investment *input*, such as R&D or SG&A expenditures, which are predominantly used in the empirical literature, might be unreliable given the considerable discretion in reporting these expenditures under current accounting standards (see footnote 3). Taking this feature of intangible investment seriously, empirical tests should focus instead on verifiable measures of investment *output*, such as, for the case of R&D investment, the number of patents granted or the market's reaction to new product launches. Indeed, considering output measures of intangible investment next to the more commonly used input measures might allow to establish more robust empirical findings also beyond the concrete predictions of our model.⁴

The multitask nature of our dynamic optimal contracting model further allows to derive new testable implications regarding the relation between pay for short-term performance and long-term investment. If, in our model, the cash flow diversion problem becomes more severe, for example due to worse corporate governance, then i) the optimal contract has to be more sensitive with respect to short-term performance. This exacerbates the agency problem with respect to long-term investment, such that, ii) investment distortions increase in absolute terms. That is, steeper short-term incentives are associated with *higher* investment expenditures in firms with high returns to investment and with *lower* investment expenditures in firms with low returns to investment.

In addition, our model provides clear predictions regarding the dependence of executive pay on the performance of intangible investment projects (see, e.g., [Balkin et al., 2000](#), for empirical evidence). Notably, our model predicts an asymmetric response of compensation to investment success and failure, with pay being more sensitive to investment success in firms with low returns to investment, where incentives are mainly provided via reward. The opposite is true in firms with high returns to investment, where incentives are mainly provided via

punishment.

The need to incentivize intangible investment creates additional volatility in the firm's cash position, which varies endogenously with the returns to intangible investment via the implemented investment profile and the pay for performance sensitivities to incentivize it. We show that the average volatility of firms' cash holdings is not monotonically related to the returns to investment, but hump-shaped as we move from industries in which successful intangible investment is less important, such as in oil, mining, or textile industries, to those in which it is highly important, such as in the pharma or the high-tech sector. This has direct implications for corporate liquidity management. In particular, our model predicts firms with intermediate returns to investment to build particularly high cash buffers before stipulating payouts (e.g., in the form of dividends). The reason is that cash holdings are particularly volatile for these firms, which exacerbates financing frictions, making precautionary cash holdings highly valuable.

Our paper contributes to the growing literature on dynamic agency problems. Optimal investment with dynamic agency is also analyzed in the two-period model of [Gertler \(1992\)](#), as well as the multiperiod discrete time models in [Quadrini \(2004\)](#); [Clementi and Hopenhayn \(2006\)](#); [DeMarzo and Fishman \(2007\)](#). Further, [DeMarzo et al. \(2012\)](#) analyze dynamic agency in a neoclassical investment setting using continuous time methods. All of these contributions consider capital investment, which can be verified by firm owners. In our model, by contrast, the manager privately controls investment and, thus has to be incentivized to invest in the firm owners' interest. Further, intangible investment in our model does not scale cash flows by increasing the capital stock as in neoclassical investment settings, but rather affects mean cash flows per unit of constant firm size.

From a methodological perspective, our model is most closely related to [Hoffmann and Pfeil \(2010\)](#), who introduce exogenous (Poisson) shocks to firm profitability in a dynamic cash flow diversion model à la [DeMarzo and Sannikov \(2006\)](#). While the firm's long-term profitability in their model is purely determined by exogenous "lucky" shocks, it can be controlled by the manager in the present analysis via investment expenditures, giving rise to a fully dynamic multitask problem (see [Holmström and Milgrom, 1991](#)). In contemporaneous work, [Hackbarth et al. \(2018\)](#), and [Gryglewicz et al. \(2020\)](#) also analyze dynamic multitask problems in which the manager controls both current cash flows via a short-term action, as well as the firm's future earnings via a long-term action. However, while the manager's long term action in their models controls firm size in the spirit of standard neoclassical capital investment, in our model of intangible investment it persistently affects future profitability per unit of constant firm size. Accordingly, the nature of observable performance signals, as a basis for incentivizing long-term investment, differs in the two settings. In our model intangible investment success and failure are rare events, corresponding, for example to the approval or rejection of a patent application, and the main mechanism driving our results is based on their relative informativeness, which depends on the implemented investment level. This mechanism is ab-

⁴The use of such output measures is already quite common in studies measuring firms' innovation efficiency, employing, in particular, the NBER patent database (see, e.g., [Hall et al., 2005](#); [Kogan et al., 2017](#)).

sent in Hackbarth et al. (2018) and Gryglewicz et al. (2020), where capital investment controls the drift of the capital stock, such that performance signals arrive continuously over time according to a diffusion process, leading to entirely different predictions regarding investment distortions. Their analysis of the agency conflicts in capital investment, thus complements ours, which captures delegated intangible investment.

Our dynamic multitask problem is also related to Zhu (2018), who considers a model of persistent moral hazard in which the agent can choose between a short-term or a long-term action in every period. He characterizes the optimal contract that always induces the long-term action. Our focus is instead on determining the optimal interior investment level at every possible history while abstracting from problems of persistent private information. Relatedly, Varas (2018) studies a multitasking model of project completion in which the agent can complete a project faster by reducing its quality, which is a one-time action. In contrast, we consider a fully dynamic repeated problem in which the agent can divert cash flow for private consumption, as well as move funds within the firm to meet either short- or long-term targets. In this sense, our model is related to the capital budgeting literature starting with Harris and Raviv (1996) and extended to a dynamic setting in Malenko (2019).

The remainder of the paper is organized as follows. We introduce the model in Section 2. In Section 3, we derive the optimal dynamic contract and provide a detailed discussion of optimal compensation and investment. In Section 4, we provide comparative statics and derive empirical implications. We conclude in Section 5. All proofs are in the Appendix.

2. The model

We consider an infinite horizon continuous time principal-agent model of a firm whose owners (the principal) hire a manager (the agent) to operate the business, which includes investment in the firm's future profitability. We first present the firm's cash flow process and investment technology. Second, we introduce the agency problem between firm owners and the manager, and formulate the optimal contracting problem.

2.1. Cash flow process and investment technology

The firm's cash flows, $\mathbf{Y} = \{Y_t : t \geq 0\}$, net of investment expenditures, $\mathbf{I} = \{I_t \in [0, \bar{I}] : t \geq 0\}$, evolve according to:

$$dY_t = (\mu_t - I_t)dt + \sigma dZ_t, \quad (1)$$

where μ_t denotes the prevailing drift rate of cash flows, which we refer to as the firm's "profitability," σ is the instantaneous volatility, and $\mathbf{Z}$ is a standard Brownian motion on a complete probability space. For notational simplicity, we restrict attention in the main text to two possible profitability levels $\mu_t \in \{\mu^l, \mu^h\}$, where $0 < \mu^l < \mu^h$. It is straightforward to extend the analysis allowing for an arbitrary number of profitability levels and our main results continue to hold in this case (see Section B of the Online Appendix).

Investment affects the firm's future profitability by determining its ability to adopt and commercialize new technologies as

they become available.⁵ As illustrated in Fig. 1, the industry is subject to (rare) exogenous technology shocks governed by a Poisson process $\mathbf{N}$ with intensity ν , indicating the availability of a new technology. If there is a technology shock ($dN_t = 1$), the firm is able to adopt the new technology with probability $p(I_t)$ in which case its future profitability will be high ($\mu_{t^+} := \lim_{s \downarrow t} \mu_s = \mu^h$). We refer to this event as an investment success, which can be interpreted as moving to or staying at the research frontier. With probability $1 - p(I_t)$, the firm is unable to adopt the new technology sufficiently quickly to stay competitive, and its future profitability will, thus, be low ($\mu_{t^+} = \mu^l$). We refer to this as an investment failure, which can be interpreted as falling or staying behind the research frontier. Note that the occurrence of an investment success is, thus, governed by a Poisson process $\mathbf{N}^s$ with arrival rate $\nu p(I_t)$ and the occurrence of an investment failure by a Poisson process $\mathbf{N}^f$ with arrival rate $\nu(1 - p(I_t))$.⁶ In between two technology shocks, profitability remains unchanged such that investment has persistent effects. Finally, we stipulate that the success probability $p(\cdot) \in [0, 1]$ is an increasing and strictly concave function of the investment amount I , satisfying $p(0) = 0$ and $p(\bar{I}) = 1$.

2.2. Agency problem

Firm owners have to hire a manager to (profitably) run the business, which relies on the manager's specific skills or private information. Both parties are risk neutral and the manager is protected by limited liability and has no initial wealth. Hence, owners have to bear the costs of setting up the firm (normalized to zero) and cover operating losses. In order to capture liquidity needs on behalf of the manager, we assume, as is standard in dynamic contracting models (e.g., DeMarzo and Duffie, 1999; DeMarzo and Sannikov, 2006), that he is relatively impatient, i.e., the discount rates of the manager (γ) and firm owners (r) satisfy $\gamma > r$.

Running the firm requires some form of discretion over the firm's cash flows, which we model as follows. Firm owners do not observe cash flows $\mathbf{Y}$ directly but only the manager's contractible reports $\hat{\mathbf{Y}}$. The difference between actual and reported cash flows is determined by the manager's hidden action, which is the source of the agency problem. In particular, we assume that the manager can divert cash flows for private consumption. To capture possible costs of concealing and taking funds out of the firm, we stipulate that the manager can consume fraction $\lambda \in (0, 1]$ of each unit diverted.

⁵For instance, investment expenditures can be interpreted as a firm's choice of absorptive capacity in the sense of Cohen and Levinthal (1990), describing its capability of assimilating new, external information and apply it to commercial ends. While we frame the investment technology within the context of technology adoption it is more widely applicable. For example, one could similarly think of the industry being subject to unpredictable demand shocks, to which the firm can react quickly enough only if it has invested sufficient resources, such as into building capacity in market research or product design.

⁶Investment in our model, thus depreciates instantly, reflecting the need to keep up with fast technological progress. Indeed, evidence suggests that R&D investment depreciates much faster than physical capital (Bernstein and Mamuneas, 2006). Nevertheless, the main drivers of our results do not depend on this assumption.

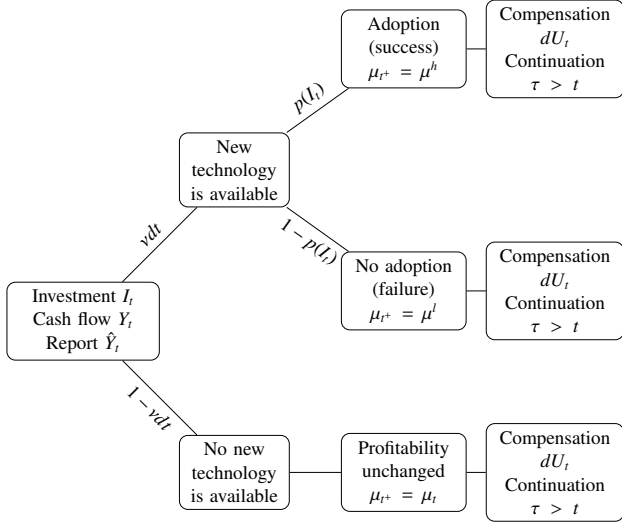


Fig. 1. Timing of events. The figure illustrates the timing of events in a generic time interval $[t, t + dt]$: i) The agent decides on investment, current cash flows are realized, and reported to the principal. ii) A new technology becomes available with probability vdt . iii) The firm is able to adopt the new technology with probability $p(I_t)$, leading to high future profitability ($\mu_{t+} = \mu^h$). With probability $1 - p(I_t)$, the firm fails to adopt the new technology and has low future profitability ($\mu_{t+} = \mu^l$). If no new technology is available (which happens with probability $1 - vdt$) profitability remains unchanged. iv) The agent gets compensated and the continuation decision is taken.

As investment similarly requires a considerable understanding of the firm's internal processes as well as the market it is operating in, it is also delegated to the manager with discretion. To model this in a concise way, we assume that, in addition to cash flows, also actual investment expenditures $\mathbf{I}$ are not contractible. Hence, the manager can freely use cash flows for investment purposes or, respectively, inflate net cash flows $\mathbf{Y}$ by deviating from the prescribed investment schedule (see (1)). While firm owners may observe (contractible) reports of $\mathbf{I}$, for example as documented in the firm's income statement, these reports do not have to coincide with actual investment expenditures, e.g., as the manager can divert funds from his investment budget for private benefit, or misrepresent running expenses as investments (see discussion in the Introduction). Hence, independently of whether or not reports of investment expenditures are available, to provide incentives, the contract has to condition on the contractible investment outcome, as governed by the Poisson processes $\mathbf{N}^g$ and $\mathbf{N}^b$.

The principal can commit to a long-term compensation contract $(\mathbf{U}, \tau)$, specifying, based on reported cash flows and investment outcomes, wage payments to the agent, dU_t , as well as the (random) time τ when the agent is fired and replaced, which incurs the principal fixed costs $k > 0$. Still, because limited liability requires the cumulative wage process $\mathbf{U} = \{U_t : 0 \leq t \leq \tau\}$ to be non-decreasing, firing and replacing the agent may be necessary for incentive provision and will indeed be part of the ex-ante optimal contract. We further assume that the agent cannot save privately, implying that $dY_t - d\hat{Y}_t \geq 0$, i.e., he can only

underreport cash flows.⁷ Hence, the agent's consumption flow at time t equals the sum of wage payments and the utility from consuming diverted funds (if any):

$$dC_t = dU_t + \lambda [dY_t - d\hat{Y}_t dt].$$

When the agent is fired, he receives his outside option, which is normalized to zero. The incumbent agent's total expected wealth at the start of his employment, hence is:

$$w_0 = E_0^S \left[\int_0^\tau e^{-\gamma t} dC_t \right], \quad (2)$$

where the expectation is conditional on time zero information and depends on the agent's reporting and investment strategy, $\mathbf{S} = \{\hat{Y}_t, I_t : 0 \leq t \leq \tau\}$.⁸ The actual value of $w_0 > 0$ in (2) is determined by the two parties' relative bargaining power. While not crucial for our results, for concreteness, we assume in what follows that the principal enjoys all bargaining power and the agent accepts any contract with $w_0 \geq 0$.

If the agent is replaced, the principal's value is given by L_τ , which denotes the (endogenous) expected profit from the relationship with a new agent, net of replacement costs k . We assume that k is sufficiently small, such that continuing the firm is optimal. Hence, at $t = 0$, the principal's total expected profit, delivering the agent an expected payoff of w_0 and given initial profitability μ_0 , is:

$$f_0 = E_0 \left[\int_0^\tau e^{-rt} (d\hat{Y}_t - dU_t) + e^{-r\tau} L_\tau \right]. \quad (3)$$

Given a compensation contract $(\mathbf{U}, \tau)$, the agent chooses a feasible strategy $\mathbf{S}$ to maximize his initial expected payoff w_0 from (2). A strategy $\mathbf{S}$ is called incentive compatible if it solves the agent's maximization problem. Hence, an incentive compatible contract can be described by the triple $(\mathbf{S}^*, \mathbf{U}, \tau)$, where $\mathbf{S}^*$ is the (incentive compatible) strategy that the principal wants to induce. The associated (global) incentive constraint is given by:

$$E_0^{S^*} \left[\int_0^\tau e^{-\gamma t} dC_t \right] \geq E_0^{\tilde{S}} \left[\int_0^\tau e^{-\gamma t} dC_t \right], \text{ for any } \tilde{S} \neq S^*. \quad (4)$$

We can simplify the analysis considerably by relying on a version of the revelation principle, which allows us to restrict attention to truth-telling contracts, implementing $\hat{\mathbf{Y}} = \mathbf{Y}$ (for a formal argument see Lemma 1 and Proposition 2 in DeMarzo and Sannikov, 2006). The contracting problem then is to find an incentive compatible truth-telling contract maximizing the principal's expected profit f_0 for given initial profitability μ_0 , delivering expected payoff w_0 to the agent and satisfying limited liability. We refer to the solution of this constrained maximization problem, which includes the optimal investment profile, as the optimal contract.

⁷Given that the agent is risk-neutral and relatively impatient, this is without loss of generality.

⁸Formally, E^S denotes the expectation under the probability measure Q^S induced by $\mathbf{S}$. If obvious, we will not state the measure associated with the expectation operator in the following.

A heuristic outline of the timing of events taking place in any infinitesimal time interval $[t, t + dt]$ prior to replacement of the incumbent manager, is illustrated in Fig. 1. Note, in particular, that the agent chooses investment prior to observing whether or not a technology shock occurred, which is the main important “sequentiality” in our continuous time model, while compensation and possible replacement of the manager occur thereafter.⁹

3. Model solution

In this section, we solve for the optimal contract using a dynamic programming approach. The optimal contract can be written in terms of the agent’s continuation payoff as the single state variable. We derive the dynamics of this key state variable along with local incentive compatibility constraints in Section 3.1. In Section 3.2, we set up the recursive formulation of the contracting problem and characterize the optimal contract. Finally, in Section 3.3, we discuss the distortions in investment relative to the first-best (owner-manager) benchmark, as well as its dynamics under the optimal contract.

3.1. Continuation payoff and local incentive compatibility

For any truth-telling contract, define the agent’s continuation payoff, w_t , as his future expected discounted payoff at time t , given he will follow strategy $\mathbf{S}^*$ from t onwards, i.e.,

$$w_t = E_t^{S^*} \left[\int_t^\tau e^{-\gamma(s-t)} dC_s \right]. \quad (5)$$

While w_t is the continuation payoff after observing whether or not a technology shock occurs in t , investment expenditures can only condition on the respective value before this uncertainty is resolved, which we denote by $w_{t-} := \lim_{s \uparrow t} w_s$. This variable will serve, together with profitability μ_t , as the state variable in the recursive formulation of the optimal contracting problem. Applying standard martingale techniques, the evolution of w_{t-} can be characterized as follows:

Lemma 1. *For any contract, there exist some predictable processes $\alpha = \{\alpha_t : 0 \leq t \leq \tau\}$ and $\beta^j = \{\beta_t^j : 0 \leq t \leq \tau\}$, $j \in \{g, b\}$, such that, if the manager follows the recommended strategy $\mathbf{S}^* = \{Y_t, I_t^* : 0 \leq t \leq \tau\}$, his continuation payoff at any moment of time evolves according to:*

$$dw_{t-} = \gamma w_{t-} dt - dU_t + \alpha_t (d\hat{Y}_t - (\mu_t - I_t^*) dt) + \beta_t^g (dN_t^g - \nu p(I_t^*) dt) + \beta_t^b (dN_t^b - \nu(1 - p(I_t^*)) dt), \quad (6)$$

where $d\hat{Y}_t - (\mu_t - I_t^*) dt$ is the increment of a Brownian motion and dN_t^g, dN_t^b are increments of standard Poisson processes with arrival rates $\nu p(I_t)$ and $\nu(1 - p(I_t))$ respectively.

⁹Formally, $\mathbf{I}$ is predictable with respect to $\mathcal{F} = \{\mathcal{F}_t, t \geq 0\}$, the filtration generated by $(\mathbf{Z}, \mathbf{N}^g, \mathbf{N}^b)$, while $\mathbf{U}$ is adapted to $\hat{\mathcal{F}} = \{\hat{\mathcal{F}}_t, t \geq 0\}$, the filtration generated by $(\hat{\mathbf{Y}}, \mathbf{N}^g, \mathbf{N}^b)$ and τ is an $\hat{\mathcal{F}}$ -measurable stopping time.

To build intuition, let us discuss in more detail the evolution of w_{t-} in (6). Due to promise keeping, the agent’s promised wealth, w_{t-} , has to grow at his discount rate, γ , while it must decrease with direct wage payments, dU_t . It also depends on his reporting strategy via sensitivity α_t and on the investment outcome via sensitivities β_t^g and β_t^b . If the agent underreports cash flows, such that he can immediately consume $\lambda(dY_t - d\hat{Y}_t)$, his continuation payoff is reduced by $\alpha_t(dY_t - d\hat{Y}_t)$. Incentive compatibility for truth-telling, thus requires $\alpha_t \geq \lambda$. Next, consider a deviation from the recommended investment I_t^* . When the agent reduces investment I_t marginally below I_t^* , from (1), net current cash flows dY_t improve at the same rate, and the agent’s continuation payoff w_{t-} grows with sensitivity α_t . However, cutting investment implies that the probability of an investment success – triggering “reward” β_t^g – decreases at rate $\nu p'(I_t^*)$, while the probability of a failure – triggering “punishment” β_t^b – increases by $\nu p'(I_t^*)$. Hence, the agent has no incentives to invest less than the recommended level $I_t^* \in (0, \bar{I}]$ if and only if $\nu p'(I_t^*) (\beta_t^g - \beta_t^b) \geq \alpha_t$. Likewise, marginally increasing investment above I_t^* increases the probability of receiving reward β_t^g by $\nu p'(I_t^*)$ and decreases the probability of punishment β_t^b by $\nu p'(I_t^*)$. However, as additional investment has to be financed out of current cash flows, the continuation payoff w_{t-} is reduced by sensitivity α_t . Hence, the agent has no incentives to invest more than the recommended level $I_t^* \in [0, \bar{I})$ if and only if $\nu p'(I_t^*) (\beta_t^g - \beta_t^b) \leq \alpha_t$. These observations lead to the local incentive compatibility conditions in Lemma 2 below.

Lemma 2. *The truth-telling contract $\{\mathbf{S}^*, \mathbf{U}, \tau\}$ with recommended investment $I_t^* \in (0, \bar{I})$ is incentive compatible if and only if*

$$\alpha_t \geq \lambda \quad (7)$$

and

$$\nu p'(I_t^*) (\beta_t^g - \beta_t^b) = \alpha_t \quad (8)$$

holds for all $t \in [0, \tau)$, almost surely.¹⁰ Further, the limited liability constraint implies for all $t \in [0, \tau)$ that

$$\beta_t^j \geq -w_{t-}, j \in \{g, b\}. \quad (9)$$

Incentive constraint (8) reflects the interaction between the problem of non-contractible investment and the cash flow diversion problem: As the agent’s continuation value must be tied to reported cash flows ($\alpha_t > 0$), it also has to increase following an investment success and decrease following a failure to provide incentives for investment. More precisely, the sensitivity of the agent’s continuation payoff with respect to the investment outcome, $\beta_t^g - \beta_t^b$, has to increase with the sensitivity to reported cash flows, α_t , for any given level of investment. Thus, a more severe cash flow diversion problem, reflected by greater diversion benefits λ , would require to impose more risk on the agent’s income in two respects. First, from (7), the sensitivity to current cash flows, α_t , has to increase, because with

¹⁰If $I_t^* = 0$, (8) is replaced by $\nu p'(I_t^*) (\beta_t^g - \beta_t^b) \leq \alpha_t$; in case $I_t^* = \bar{I}$, we instead require $\nu p'(I_t^*) (\beta_t^g - \beta_t^b) \geq \alpha_t$.

higher diversion benefits, the agent's income has to be linked more closely to reported cash flows to induce truthful reporting. Second, with his income being more sensitive to current cash flows, the agent has an incentive to inflate those by deviating from the recommended investment level. Hence, according to (8), his income has to be more responsive to the investment outcome, i.e., to future profitability as well.

3.2. Optimal contract

The optimal contract can now be derived using the dynamic programming approach. Denote by $f^i(w)$, $i \in \{l, h\}$ the principal's value function, that is, the highest profit the principal can attain under any incentive compatible contract delivering expected payoff w to the agent and given the prevailing drift rate of cash flows, μ^i , $i \in \{l, h\}$, where we drop the time subscripts for notational convenience.

The contracting problem can be greatly simplified by noting that, following any history, the optimal (continuation) contract is independent of the current profitability state μ^i . Intuitively this result follows since, for a given level of investment, the probability of an investment success or failure does not depend on current profitability, nor do the basic cash flow diversion problem or the agency problem with respect to investment. As the optimal compensation and investment policies are, thus independent of the prevailing profitability level, the principal's value in the two profitability states only differs by an additive constant, which captures the direct benefit from being at the research frontier, $\mu^h - \mu^l$, properly accounting for the Markov-switching structure.

Lemma 3. *The optimal contract (S^*, U, τ) is independent of current profitability μ^i and the principal's value functions in the high and in the low profitability state satisfy*

$$f(w) := f^l(w) = f^h(w) - \Delta, \quad (10)$$

with

$$\Delta := \frac{1}{r + \nu} (\mu^h - \mu^l). \quad (11)$$

From Lemma 3 we can now characterize the optimal contract based on the agent's continuation payoff w as the single state variable. Consider first the optimal compensation policy. Clearly, the principal can always compensate the agent directly by paying him a lumpsum of $dU > 0$ (at marginal costs of -1) and then move to the optimal contract with reduced continuation payoff $(w - dU)$. However, deferring compensation may be valuable. A higher promised wealth, w , relaxes future limited liability constraints as the agent can be punished for poor results by "clawing-back" previously deferred payments instead of having to inefficiently fire and replace him at cost k . In contrast to the costs of deferring compensation, which are due to the wedge in discount rates ($\gamma > r$), this benefit declines, however, as the agent's continuation payoff, w , increases and with it the probability of inefficient firing. This is reflected in the concavity of $f(w)$, which we show formally in the Appendix (see proof of Proposition 1). As a consequence, the principal is effectively risk-averse with respect to variation in

the agent's continuation value. One implication is that compensation is optimally deferred until a threshold $\bar{w}$ is reached, where $f'(\bar{w}) = -1$ and the agent is paid cash.

Next, since the principal discounts at rate r , his expected flow of value at time t must be $rf^i(w)dt$. This has to be equal to the expected instantaneous net cash flow $(\mu^i - I)dt$ plus the expected change in his value function, which can be computed using Itô's lemma and the change of variables formula for jump processes. Hence, on the interval without cash compensation, $w \in [0, \bar{w}]$, the principal's value function in the low profitability state must satisfy the following HJB equation ($f^h(w)$ then follows from (10)):

$$\begin{aligned} (r + \nu) f(w) = \max_{\alpha, \beta^g, \beta^b, I} & \left\{ \mu^l - I + \nu p(I) \Delta + \frac{1}{2} \sigma^2 \alpha^2 f''(w) \right. \\ & + [\gamma w - \nu [\beta^g p(I) + \beta^b (1 - p(I))]] f'(w) \\ & \left. + \nu p(I) f(w + \beta^g) + \nu (1 - p(I)) f(w + \beta^b) \right\} \end{aligned} \quad (12)$$

s.t. (7), (8), (9).

Since cash transfers cause w to reflect at $\bar{w}$, the value function extends linearly to the right of the compensation threshold, i.e., $f(w) = f(\bar{w}) - (w - \bar{w})$ for $w \geq \bar{w}$. To solve for the optimal contract, we have to pin down a solution to (12) and the compensation threshold $\bar{w}$. For this we require three boundary conditions. The first ("value matching") condition,

$$f(0) = f(w^*) - k, \quad (13)$$

with $w^* \in \arg \max_w \{f(w)\}$, reflects that upon firing the incumbent agent at $w = 0$, the principal receives $f(w^*)$ from the relation with a new agent and bears replacement costs k . The second condition is a standard "smooth pasting" condition at the compensation threshold:

$$f'(\bar{w}) = -1, \quad (14)$$

while the third boundary condition ("super contact") guarantees the optimal choice of $\bar{w}$:

$$f''(\bar{w}) = 0. \quad (15)$$

In Proposition 1 we characterize the optimal contract (i.e., the solution to the boundary value problem in (12)-(15)).

Proposition 1. *The optimal truth-telling contract with non-contractible investment takes the following form: Optimal investment $I(w)$ as well as the sensitivities $\alpha(w) = \lambda$ and $\beta^j(w)$, $j \in \{g, b\}$, are independent of μ^i , $i \in \{l, h\}$, and chosen as maximizers in (12). The incumbent agent's continuation payoff evolves according to (6) with $\alpha_t = \lambda$, $\beta_t^j = \beta^j(w_t)$, $j \in \{g, b\}$ and $I_t = I(w_t)$, $\forall t$. Optimal compensation satisfies $dU_t = \max\{w_t - \bar{w}, 0\}$. The incumbent agent is replaced when $w_t = 0$. The principal's expected payoff at any point in time is given by $f^i(w_t)$, $i \in \{h, l\}$, which satisfies $f(w) := f^l(w) = f^h(w) - \Delta$, where $f(w)$ is concave, strictly so for $0 \leq w < \bar{w}$ and solves for $w \in [0, \bar{w}]$, the HJB equation in (12), subject to the boundary conditions (13) to (15).*

As is standard in dynamic cash flow diversion models (e.g., DeMarzo and Sannikov, 2006), the agent's incentive constraint with respect to cash flow diversion (7) binds under the optimal contract. Formally, this result follows from the concavity of $f(w)$ and the observation that increasing α would increase the instantaneous volatility of w . Given that the basic cash flow diversion problem is by now well understood, in Section 3.3, we focus on the investment task and discuss the optimal investment schedule, as well as the optimal choice of reward and punishment used to incentivize investment under the optimal contract of Proposition 1.

3.3. Optimal investment

Incentive compatibility requires that the optimal contract contains a certain degree of reward following an investment success, $\beta^g(w) > 0$, and/or punishment after a failure, $\beta^b(w) < 0$. However, providing these incentives for investment is costly such that, in determining the optimal level of investment, the principal trades off the potential for higher profitability with the additional agency costs of providing incentives for investment. To see this formally, note that from (12) we can write the principal's problem of finding the optimal investment level as follows:

$$I(w) \in \arg \max_{I \in [0, \bar{I}]} \{vp(I)\Delta - I - \Phi(w, I)\}, \quad (16)$$

with

$$\begin{aligned} \Phi(w, I) = & \min_{\beta^g, \beta^b \text{ s.t. (8),(9)}} \left\{ vp(I)[f(w) + \beta^g f'(w) - f(w + \beta^g)] \right. \\ & \left. + v(1 - p(I))[f(w) + \beta^b f'(w) - f(w + \beta^b)] \right\}. \end{aligned} \quad (17)$$

From (16), optimal investment is chosen to maximize the (expected) technological returns to investment, $vp(I)\Delta$, net of investment expenditures, I , and the agency costs of providing incentives for investment denoted by Φ . Since firm owners are effectively risk-averse with respect to variation in the manager's compensation, as reflected in the concavity of $f(w)$, these agency costs are strictly positive whenever incentive compatible rewards for investment success, β^g , and punishments for investment failure, β^b , induce additional variation in the agent's continuation payoff w . Accordingly, the optimal combination of reward and punishment to implement a given level of investment are determined from (17), minimizing the agency costs of providing incentives for investment, Φ .

We next examine optimal compensation and investment distortions for a given value of the agent's continuation utility w . Then, we analyze the implied compensation and investment dynamics when w evolves as described in Proposition 1.

3.3.1. Optimal incentives for investment

We, first, derive the optimal compensation policy implementing a given level of investment at lowest agency costs. So minimizing Φ as given in (17) with Lagrange multiplier $\eta(w)$ attached to the incentive compatibility constraint (8), implies that

with slack limited liability (9) interior optimal values $\beta^g(w)$ and $\beta^b(w)$ have to satisfy

$$[f'(w) - f'(w + \beta^i(w))] / LR^i(I(w)) = \eta(w), \quad (18)$$

for $i \in \{g, b\}$. Here $LR^i(I) := \frac{d}{dI} \log(\Pr(\mu = \mu^i | I))$ denotes the likelihood ratio of the respective investment outcome, i.e., $LR^g(I) = p'(I)/p(I)$ for an investment success and $LR^b(I) = -p'(I)/(1 - p(I))$ for an investment failure. As is well known, $|LR^i(I)|$ measures the strength of incentive provision per unit of expected reward or punishment and, in that sense, how informative a performance signal — here, an investment outcome — is for providing incentives (e.g., Holmström, 1979). Intuitively, the numerator on the left-hand side in (18) captures the marginal cost of providing incentives through reward or punishment arising from the principal's "risk aversion," while the denominator captures how effective the respective transfer is for providing incentives. The optimal compensation policy is then chosen such as to equalize these cost-benefit ratios of providing incentives through reward and punishment.

To build further intuition, consider the benchmark case with contractible investment (henceforth, CI ; see Section A of the Online Appendix for a complete characterization). Then, as the contract does not have to provide incentives for investment, optimal risk-sharing is attained by setting $\beta_{CI}^g(w) = \beta_{CI}^b(w) = 0$. Thus the principal, who is effectively risk-averse, bears all risk from the investment shock and perfectly insures the risk-neutral agent. This seemingly counterintuitive result follows from the fact that the principal is risk-averse with respect to variation not in his own, but in the agent's income, such that optimal risk sharing (corresponding to $\eta(w) = 0$ in (18)) prevails whenever the agent's compensation is not contingent on investment outcomes that occur with positive probability.

Instead, with non-contractible investment, the constrained optimal solution, subject to incentive compatibility, will usually have to deviate from optimal risk-sharing. According to (18), this is optimally done by shifting most of the required incentive costs to the more informative investment outcome, i.e., by setting

$$\frac{f'(w + \beta^b(w)) - f'(w)}{f'(w) - f'(w + \beta^g(w))} = \left| \frac{LR^b(I(w))}{LR^g(I(w))} \right|. \quad (19)$$

With binary investment outcomes, the relative likelihood ratio can be conveniently written as $|LR^b(I)/LR^g(I)| = p(I)/(1 - p(I))$, such that from (19) incentive costs are mainly incurred following the investment outcome that is less likely to occur. Now, as the probability of successful investment increases in the implemented investment level, we obtain the result in Lemma 4, comparing the composition of incentive costs for different sets of parameters $\psi \in \{\mu^h, \mu^l, v, \lambda, r, \gamma, \sigma, k\}$, which for given w map into different $I(w)$.

Lemma 4. Fix $w \in (0, \bar{w})$ and consider two sets of parameter values ψ' and ψ'' , such that $I(w)_{|\psi'} < I(w)_{|\psi''}$ under the optimal contract of Proposition 1. Then, as long as the limited liability constraint does not bind, rewards for investment success, $\beta^g(w)$, and punishments for investment failure, $\beta^b(w)$, are

chosen such that the ratio of the costs of providing incentives through punishment relative to the costs of providing incentives through reward is strictly increasing in $I(w)|_\psi$, i.e.,

$$\left. \frac{f'(w + \beta^b(w)) - f'(w)}{f'(w) - f'(w + \beta^g(w))} \right|_\psi < \left. \frac{f'(w + \beta^b(w)) - f'(w)}{f'(w) - f'(w + \beta^g(w))} \right|_{\psi'}.$$

Further, $\beta^g(w) \rightarrow 0$ as $I(w)|_\psi \rightarrow \bar{I}$, while for $I(w)|_\psi \rightarrow 0$, as the agency problem vanishes, $-\beta^b(w)/\beta^g(w) \rightarrow 0$, i.e., the first unit of investment is incentivized with rewards only. For all $I(w)|_\psi \notin \{0, \bar{I}\}$, both reward and punishment are used, i.e., $\beta^g(w)|_\psi > 0 > \beta^b(w)|_\psi$.

This dependence of the relative costs of providing incentives through punishment rather than through reward on the implemented level of investment will be key in understanding the distortions in investment, which we analyze next.

3.3.2. Investment distortions

In the first-best benchmark without agency frictions, a profit-maximizing risk-neutral owner-manager with discount rate r would choose the following investment level for all $t \geq 0$:

$$I_{FB} \in \arg \max_{I \in [0, \bar{I}]} \{vp(I)\Delta - I\}. \quad (20)$$

This first-best investment schedule is constant and satisfies, if interior, the first-order condition $vp'(I)\Delta = 1$, which trades off the potential for higher profitability against the marginal investment expenditures. Notably, first-best investment also obtains in the presence of the cash flow diversion problem as long as investment is contractible (see Section A of the Online Appendix for details). This is due to the fact that both the expected returns to investment, captured by $vp(I)\Delta$, as well as its costs, I , are independent of the agency problem.

In order to determine whether, with non-contractible investment expenditures, the firm invests too much or too little relative to this (first-best) benchmark, it is crucial to understand how the agency costs of investment in (17) change in I , i.e., to sign the marginal agency costs of investment $\partial\Phi(w, I(w))/\partial I$. If these are positive, investment is distorted downwards relative to first-best, while it is distorted upwards if marginal agency costs are negative. We next show that agency costs of delegated investment are non-monotonic in the incentivized investment level I such that indeed both under- and overinvestment can arise in equilibrium.

To see this, it is instructive to look first at the boundary cases where the problem in (20) has a corner solution. When the investment technology is very unprofitable, such that, in the extreme case, $I_{FB} = 0$, the principal can trivially implement the first-best investment level without incurring any agency costs as there is no need to provide incentives. Hence, optimal risk-sharing can be attained. However, also when the investment technology is highly profitable, such that $I_{FB} = \bar{I}$, first-best investment can be incentivized without inducing (additional) volatility in w (i.e., maintaining optimal risk-sharing). Now, the shadow value on the incentive constraint in (18) is zero as an investment failure becomes extremely informative, in the sense

of the respective likelihood ratio going to infinity. Intuitively, when the probability of success is $p(\bar{I}) = 1$, from Lemma 4, all incentives are optimally provided “off-equilibrium” by relying on punishment only, as long as this does not violate limited liability. Since agency costs, as given in (17), are zero for $\beta^g(w) = 0$ and $p(\bar{I}) = 1$, optimal risk-sharing prevails and optimal investment equals the first-best level.

Lemma 5. Fix $w \in [0, \bar{w}]$. Then, as long as the limited liability constraint does not bind, investment under the optimal contract of Proposition 1 is equal to first-best whenever first-best investment attains a corner solution, i.e., $I(w) = I_{FB}$ for $I_{FB} \in \{0, \bar{I}\}$.

The discussion above makes clear that first-best investment can be implemented without incurring any agency costs if $I_{FB} = 0$, or $I_{FB} = \bar{I}$. However, this is no longer true for interior values of I_{FB} which, from Lemma 4, are optimally implemented using both reward, $\beta^g > 0$, as well as punishment, $\beta^b < 0$, thus implying a deviation from optimal risk-sharing. Since agency costs to incentivize first-best investment are strictly positive if I_{FB} is interior, and zero when I_{FB} attains a corner solution, it is easy to see that marginal agency costs are positive (negative) if first-best investment is sufficiently low (high). As a result, investment under the optimal contract will be distorted downwards in the former and upwards in the latter case.

Proposition 2. Fix $w \in (0, \bar{w})$. Then, as long as the limited liability constraint does not bind, investment is distorted downwards, $I(w) < I_{FB}$, for sufficiently low values of $I_{FB} > 0$, while it is distorted upwards, $I(w) > I_{FB}$, for sufficiently high values of $I_{FB} < \bar{I}$.

To build intuition, note that increasing investment has two basic effects on agency costs. First, higher investment requires stronger incentives (see (8)), and, second, it affects the outcome distribution, thus, increasing the probability of having to reward the agent for a favorable investment outcome and decreasing the probability of having to punish him for an investment failure. While the first effect unambiguously tends to increase agency costs pushing towards underinvestment, the second effect can have either sign depending on the relative costs of providing incentives through reward rather than punishment. As Proposition 2 shows, overinvestment may, hence, arise if incentives are predominantly given through punishment for bad outcomes, which from Lemma 4 is the case if the investment technology is sufficiently profitable such that an (unlikely) investment failure constitutes a very informative performance signal as measured by a high relative likelihood ratio. Then, to reduce the probability of having to bear the high agency costs associated with an investment failure, it is optimal to increase $I(w)$ above the first-best value. The opposite case applies if the investment technology is rather unprofitable, implying low first-best investment, which would be optimally incentivized with high rewards following an informative investment success. Then, to avoid the associated high incentive costs, investment $I(w)$ is optimally distorted downwards relative to first-best.

As Proposition 2 illustrates, the sign of the investment distortion crucially depends on the profitability of the investment

technology as captured by I_{FB} . Now, in order to clearly separate the technological profitability of investment from the associated agency problem, going forward, we will vary I_{FB} by changing the respective “returns to investment” $\Delta\mu := \mu^h - \mu^l$ which is a purely technological parameter that — different from other parameters determining I_{FB} — does not directly affect the agency problem.¹¹ As the following example illustrates, ranking investment technologies according to these “returns to investment” allows us to determine a cutoff below which investment is distorted downwards and above which investment is distorted upwards relative to first-best.

Example 1. Consider $p(I) = \phi \sqrt{I}$, and fix parameters $\nu = 1.2$, $\sigma = 10$, $\lambda = 0.5$, $k = 15$, $\gamma = 0.15$, $r = 0.1$, and $\phi = 0.95$. Then the manager overinvests under the optimal contract (for all $w \in (0, \bar{w})$ for which the limited liability constraint is slack), if “returns to investment” $\Delta\mu$ are larger than a cutoff $\Delta\mu^c = 1.77$, while he underinvests for $\Delta\mu < 1.77$. Given the other parameter values, this corresponds to a cutoff in first-best investment level of $I_{FB}^c = 0.6$ or a success probability of $p(I_{FB}^c) = 0.74$.

3.3.3. Compensation and investment dynamics

So far, we have studied how investment distortions and the incentive scheme used to implement investment depend on the profitability of the investment technology as captured by I_{FB} . For this analysis, we kept the agent’s continuation value fixed. We next analyze the resulting dynamics of investment and compensation when w evolves as described in Proposition 1. In order to do so, it is useful to look first at extreme values of the state space, i.e., $w = \bar{w}$ and $w = 0$.

As the agent’s continuation value reaches the compensation boundary ($w = \bar{w}$), he has accumulated so much wealth inside the firm that the agency problem is relaxed sufficiently for firm owners’ effective risk-aversion to disappear. In fact, rewards are taken out as cash payments and do not induce costly variation in w (see Proposition 1). Hence, it is optimal not to punish the agent and to rely exclusively on rewards to incentivize investment. As a result, agency costs of investment are zero and investment equals first-best.

For $w < \bar{w}$, the principal’s value function is strictly concave such that both rewards and punishments are used according to the optimal compensation policy in (19). However, for sufficiently low values of the agent’s continuation payoff, punishment is restricted by the limited liability constraint (9). In fact, only rewards can be used to incentivize any $I > 0$ in the limiting case as we approach the termination boundary ($w = 0$) and the agent is “too poor to be punished.” Now, whenever incentives are provided mainly through rewards, an investment success is the investment outcome that is particularly costly in terms of incentive provision. As a result, investment will be optimally distorted downwards for all $I_{FB} \in (0, \bar{I})$. In Lemma 6, we summarize these observations.

Lemma 6. Fix $I_{FB} \in (0, \bar{I})$. Then under the optimal contract of Proposition 1:

- i) For $w \geq 0$ sufficiently small, the limited liability constraint for $\beta^b(w)$ binds such that the agent is instantly fired following a failure. Investment is distorted downwards, $I(w) < I_{FB}$.
- ii) For $w < \bar{w}$ sufficiently large, incentives are provided through punishment and reward, which are optimally chosen according to (19). Investment may be distorted upwards or downwards as shown in Proposition 2.
- iii) For $w \geq \bar{w}$, incentives are provided through rewards only and investment is equal to first-best, $I(w) = I_{FB}$.

Fig. 2 illustrates the compensation and investment dynamics implied by Lemma 6 in an equilibrium with overinvestment (high I_{FB} in left panels), and in one with underinvestment (low I_{FB} in right panels). As discussed above, the two scenarios are generated by varying the “returns to investment” $\Delta\mu := \mu^h - \mu^l$, while all other parameters and functional forms are kept constant. As shown in Lemma 6, there is no punishment ($\beta^b(w) = 0$) if the agent’s track record is very poor ($w \rightarrow 0$) or if it is very good ($w \rightarrow \bar{w}$), while $\beta^b(w)$ is strictly negative in between. The upper panels of Fig. 2 illustrate cases where the punishment policy $\beta^b(w) \leq 0$ is in fact U-shaped in the agent’s continuation value w . The lower the degree of punishment, keeping all else constant, the more generous rewards are needed to satisfy incentive compatibility. Notably, this is true independently of whether the lack of punishment is imposed by the binding limited liability constraint (for low w), or if it is in fact optimal not to punish (for high w). As a result, also the reward policy $\beta^g(w) \geq 0$ in our numerical example is U-shaped in w .

Turning to the optimal investment policy $I(w)$ as depicted in the lower panels of Fig. 2, note first that investment is distorted downwards relative to first-best for low returns to investment $\Delta\mu$ (right panel) and upwards for high $\Delta\mu$ (left panel), as long as w is sufficiently large, which corresponds to a slack limited liability constraint. In both cases, investment $I(w)$ approaches the respective first-best value at the compensation boundary, i.e., for $w \rightarrow \bar{w}$. Hence, in the case with high $\Delta\mu$ (left panel), investment is distorted upwards and decreases with the agent’s stake in the firm w , while for low $\Delta\mu$ (right panel), it is distorted downwards and increases with w . In line with empirical evidence, our model, thus, predicts positive investment-cash flow sensitivities in firms with low returns to investment and (excluding financially distressed firms) negative investment-cash flow sensitivities in firms with high returns to investment.

4. Empirical implications

In this section, we use our dynamic agency model with non-contractible investment to derive empirical implications for the level and dynamics of investment in intangibles, the structure of executive compensation, liquidity management, and payout policies. Before moving to these implications, we have to map key model variables and parameters into observables.

¹¹I.e., for a fixed (exogenously given) investment schedule, the optimal contract would be the same for all values of $\Delta\mu$.

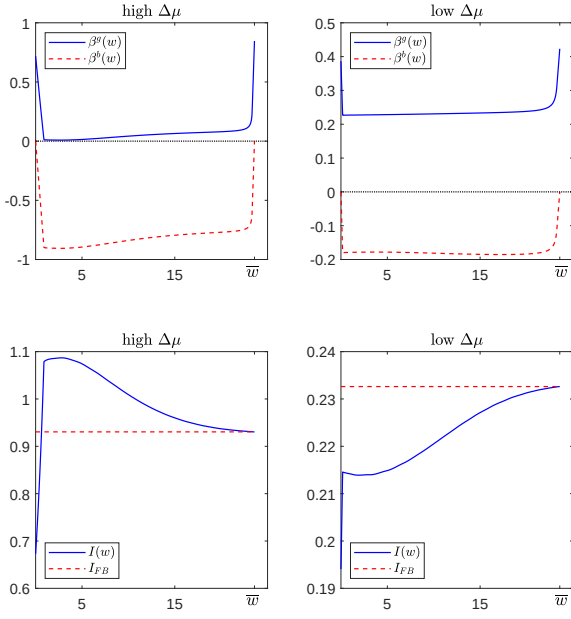


Fig. 2. Compensation and investment dynamics. The upper panels illustrate the punishment and reward dynamics and the lower panels the investment dynamics under the optimal contract for low $\Delta\mu = 1.1$ and for high $\Delta\mu = 2.2$. All remaining parameters are set as in Example 1.

4.1. Implementation and measurement of key model variables

4.1.1. Input- vs. output-based measures of investment

In order to test our model's predictions regarding investment, we have to find reliable empirical measures of the non-contractible investment expenditures we seek to model. Recall that by focusing on non-contractible investment in future profitability per unit of (physical) capital instead of standard capital investment, our model seeks to explain intangible investment, such as investment in knowledge or organizational capital. Common input proxies for these types of intangible investment include R&D expenditures or parts of selling, general and administrative (SG&A) spending (e.g., [Peters and Taylor, 2017](#)). However, within these applications, actual and reported investment expenditures need not coincide perfectly. For instance, while R&D investments have to be expensed and reported when material ([FASB, 1974](#)), accounting rules such as U.S. Generally Accepted Accounting Principles (GAAP) do leave considerable discretion to managers regarding which costs to classify as R&D.¹² Exploiting this discretion (e.g., to artificially improve short-term “core” earnings at the cost of long-term profitability) is referred to as “classification shifting” in the accounting literature (see discussion in Introduction).

¹²The Financial Accounting Standards Board ([FASB, 1974](#)) acknowledges that “the differences [in research and development] among enterprises and industries are so great that a detailed prescription of the activities and related costs includable in research and development, either for all companies or on an industry-by industry basis, is not a realistic undertaking for the FASB.” See also [Horwitz and Kolodny \(1980\)](#).

This is similarly relevant for investment in organizational capital, including human capital, brands, customer relationships, and distribution system, for which only relatively coarse measures (e.g., in the form of SG&A) are available.

These examples illustrate the relevance of our key agency friction, the non-observability of actual investment expenditures, for different types of intangible investment. Our theoretical analysis focuses on the optimal dynamic investment incentives in such settings in which reported investment expenditures do not need to coincide with actual investment. Under the optimal contract, managers invest in the firm owners' interest, which uniquely pins down actual investment. Reported investment expenditures on the other hand affect neither the manager's nor investors' value and, hence, are not uniquely determined under the optimal contract. In fact, the manager and investors could agree on any “reporting convention” not necessarily just truthtelling. The possibility of classification shifting (i.e., misreporting) in equilibrium poses a challenge for any empirical test taking reported investment expenditures as a measure of actual intangible investment. In light of our model, empirical tests of firms' actual investment in intangibles such as R&D should thus be based on observable measures of investment output rather than on direct but possibly manipulated input measures in the form of reported investment expenditures.

Accordingly, all subsequent empirical implications are stated in terms of two measures of investment output, which are both observable and pinned down uniquely by the optimal contract. Investment output in our theory model comes in the form of investment successes and failures. The former become more, the latter less frequent, as actual investment expenditures increase according to the respective arrival rates of $\nu p(I)$ and $\nu(1 - p(I))$. The first measure of investment output we use is the success rate, which is the number of observed successes per unit of time.¹³ In the context of our main application to R&D, this could be measured empirically by the number of patents filed or a count of successful product launches within each year. While the success rate essentially is a “counting measure,” our second measure of investment output, firm profitability, μ , further captures the value of an investment success or failure. It corresponds for instance to a gain in market share, whenever the respective firm is at the research frontier, while falling behind the research frontier results in respective losses (see, e.g., [Kogan et al., 2017](#); [Hall et al., 2005](#), for empirical estimates of the market value of innovative output).

4.1.2. Implementation and cross-sectional averages

In order to derive testable implications from our dynamic agency model, we next show how to interpret key variables and parameters of the optimal contract within an implementation.

While it is well known that such an implementation is not unique (see, e.g., [DeMarzo et al., 2012](#), for a discussion), the robust feature shared by all common implementations is that the key state variable, the agent's continuation payoff, is interpreted

¹³While we state the subsequent implications in terms of the success rate, they clearly also hold — with reverse sign — for the respective failure rate.

as a measure of the firm's financial slack. To see this relation, recall that when the agent's continuation payoff falls to zero, this triggers a restructuring of the firm, which is costly for its owners and involves the replacement of the incumbent management. As, for incentive reasons, the agent's continuation payoff w_t moves with cash flows with sensitivity λ , the maximum loss that a firm can sustain without triggering a restructuring, i.e., its financial slack, is equal to $M_t := w_t/\lambda$. While this insight can be formalized in a variety of ways, we consider for illustrative purposes an implementation of the optimal contract based on cash reserves as a measure of financial slack (see Section C of the Online Appendix for details). When cash holdings reach the payout boundary, $M^{pb} := \bar{w}/\lambda$, the firm issues dividend payments to its shareholders and, thus, also the manager who holds fraction λ of the firm's equity. If cash reserves fall to zero, which could be interpreted as insolvency, the firm needs to raise cash from the capital market, which involves a fixed cost of k . These costs of raising external funds capture the key financing friction in this implementation of the optimal contract. In the following, we therefore interpret all state-dependent variables as a function of cash holdings, e.g., for the case of investment, $I_t := I(\lambda M_t)$, where $M_t = w_t/\lambda$.

Finally, in order to derive cross-sectional implications from our dynamic single-firm model, we need to compute the cross-sectional distribution of cash holdings M . For this, note that since the firm in our model operates forever, i.e., within the implementation, it is always optimal to refinance it when internal funds are depleted, the stationary (cross-sectional) distribution of the state variable M can be computed by simulating a sufficiently long path for a single firm.¹⁴ We then use the stationary distribution of $M = w/\lambda$ to compute cross-sectional averages (denoted by a bar, e.g. for the case of investment, $\bar{I}$).

In the theoretical analysis in Section 3.3, we discuss the distortions in and the dynamics of actual investment expenditures as a function of the agent's continuation value, corresponding to the firm's cash holdings within our implementation. In particular, we show that actual investment expenditures are distorted upwards and decrease in cash holdings in firms with very profitable investment technologies (as captured by high first-best investment), while with less profitable investment technologies, investment is distorted downwards and increases in cash holdings. In the same spirit, we formulate the empirical implications below in terms of cross-sectional averages of firms for which intangible investment is more important, such as in the pharmaceutical or the high-tech sector, and those for which it is less important, such as in the oil, mining, or textile industries. For concreteness, we parameterize the importance of intangible investment by varying the sensitivity of expected cash flows to the investment outcome, i.e., $\Delta\mu = \mu^h - \mu^l$ (see also the discussion

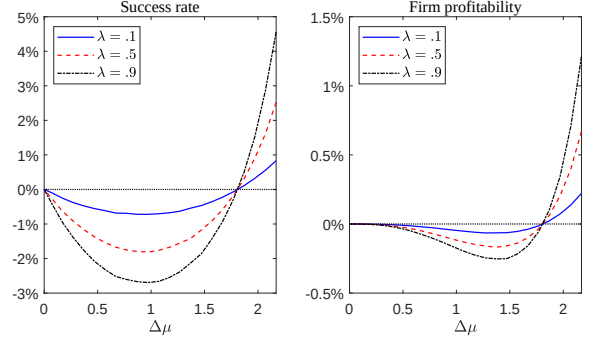


Fig. 3. Distortions relative to owner-manager benchmark. Plots percentage distortions of the average success rate (left panel) and the average firm profitability (right panel) relative to the owner-manager benchmark, as a function of returns to investment $\Delta\mu$, which we vary between 0 and an upper limit that guarantees an interior solution for the optimal investment level solving (16). All remaining parameters are set as in Example 1.

surrounding Example 1 above).

4.2. Implications for investment distortions and dynamics

In this section, we derive testable implications regarding the efficiency of investment, its dependence on the firm's cash holdings, and how this relation is affected by financial frictions. We also consider the relation between corporate governance, short-term compensation, and investment. Throughout we focus on cross-sectional implications for observable measures of investment output (success rate and profitability) as introduced above.

4.2.1. Distortions in intangible investment.

Since (unobservable) investment input is monotonically related to investment output, as measured by either the success rate or profitability, it is immediate that they follow a similar pattern, which allows for empirical testing of investment distortions. In particular, comparing average investment output when investment input is delegated (and non-contractible) with the “owner-manager” (or contractible investment) benchmark, we obtain:

Implication 1. *Relative to comparable owner-manager firms, average investment output under delegation is distorted*

- i) *downwards when returns to investment are low, and*
- ii) *upwards when returns to investment are high.*

In Fig. 3, we plot the percentage deviation of average success rates (left panel) and average profitability (right panel) from the owner-manager benchmark as a function of the returns to investment $\Delta\mu$. When $\Delta\mu$ is low, the average success rate is lower than that of comparable owner-manager firms, which corresponds for example to a lower patent count or fewer innovations in general. On the other hand, in firms with high returns to investment, the average success rate is higher than in comparable owner-manager firms (left panel). The same qualitative pattern of distortion holds for our valuation measure of investment

¹⁴That is, cross-sectional (“ensemble”) averages can be obtained by computing the corresponding path-wise long-term averages, e.g., $\bar{M} := \frac{1}{T} \sum_{t=0}^T M_t$, with T sufficiently large. This ergodicity property is exploited also by Brunnermeier and Sannikov (2014) and DeNicoló et al. (2021). We compute cross sectional averages in this way by Monte-Carlo simulation with a total number of $T = 10^7$ periods with period length $h = 10^{-3}$. We further specify $p(I) = \phi \sqrt{I}$, with $\phi > 0$ such that $I \in [0, 1/\phi^2]$. For more details and an outline of the employed algorithm, we refer to Section D of the Online Appendix.

output, namely firm profitability (see right panel). Notably, the distortions in investment (output) depicted in Fig. 3 arise endogenously via the optimal contractual response to a multitask problem, without assuming any direct costs or benefits of investment on the manager's side, such as empire building preferences or pressure from the stock-market.

4.2.2. Pay for short-term performance, corporate governance, and investment output

A common presumption in much of the literature on managerial short-termism is that if pay is more sensitive to short-term performance, managers will be biased towards improving the former, while neglecting (hard to verify) long-term investment. This would then lead to lower investment output in the long-run (e.g., [Bebchuk and Fried, 2004](#); [Edmans et al., 2017](#)) Fig. 3 illustrates that, under a long-term contract that optimally balances incentives over different horizons, the relation between pay for short-term performance and long-term investment output can in fact be positive. In particular, as the agent's diversion benefit λ increases, more compensation for short-term performance is needed to induce truth-telling according to incentive constraint (7). By incentive constraint (8), then also the exposure to investment output needs to increase to incentivize any given level of investment input. As a consequence, investment distortions are aggravated, which implies more severe overinvestment (underinvestment) in firms with high (low) returns to investment, $\Delta\mu$.

Implication 2. *The relation between investment output and the sensitivity of pay to short-term performance is:*

- i) *negative in firms with low returns to investment, and*
- ii) *positive in firms with high returns to investment.*

Given that better corporate governance reduces the manager's potential to divert funds for private consumption (lower λ), Implication 2 can also be interpreted as a statement on the relation between corporate governance and investment output.¹⁵ Our multitask model therefore describes a novel channel, relating the impact of corporate governance on investment output to industry characteristics, which does not assume any direct costs or benefits of investment on the manager's side.

4.2.3. Investment-cash sensitivities

So far, we have considered distortions in investment output and associated comparative statics for a given level of cash holdings. Next, we derive implications for the dynamics of investment, in particular its sensitivity to cash flows. In doing so,

we again focus on the two observable measures of investment output.

In Fig. 4, we plot the average elasticity of investment output with respect to cash holdings implied by the optimal contract as a function of the returns to investment. Since higher cash flows ceteris paribus imply higher cash holdings, the sensitivity of investment to cash flows follows a similar pattern. To understand the depicted pattern, note that, intuitively, the agency problem is mitigated when the manager's "stake" in the firm increases. As a consequence, investment distortions relative to the constant first-best level decrease with positive firm performance (i.e., with higher cash flows), and, thus, higher cash holdings. Hence, average investment cash (flow) sensitivities have the opposite sign as the average investment distortions in Implication 1.

Implication 3. *The sensitivity of investment output to internal funds (cash) is:*

- i) *positive in firms with low returns to investment, and*
- ii) *negative in firms with high returns to investment.*

There is a large empirical literature (starting with [Fazzari et al., 1988](#)) that analyzes the relation between firms' investment and cash flows. Initially, studies focused on capital investment, although recent papers have explicitly considered intangible investment, such as R&D spending, showing that the distortions and dynamics of intangible investment turn out to differ fundamentally from that of capital investment. For instance, [Peters and Taylor \(2017\)](#) find small, or even negative investment-cash flow sensitivities in industries with a high intensity of intangible investment, such as the high-tech and health sectors (see also [Hovakimian, 2009](#)). Instead, they document a positive sensitivity in industries with a low intangible investment intensity, such as the consumer and the manufacturing sectors. Hence, while they use reported investment input to measure intangible investment, their results are in line with our model's predictions regarding investment output in Implication 3. Given the growing importance of intangible investment, the finding that it may both increase and decrease in cash flows, depending on firms' returns to such investment, may further explain recent empirical findings documenting the decline or even disappearance of investment-cash flow sensitivities in a broad cross-section of firms over the last decades (see, e.g., [Brown and Petersen, 2009](#); [Chen and Chen, 2012](#)).

4.2.4. Financial constraints

Much of the empirical investment-cash flow sensitivity literature has evolved around the question of whether these sensitivities are a good proxy for financial constraints. While this literature has focussed mostly on capital investment, our model's predictions regarding the dependence of intangible investment on cash flow suggest a novel perspective on this debate. We find that more severe financial constraints as captured by higher costs of external financing, k , imply a higher sensitivity of intangible investment (output) to cash holdings and, thus, cash flows. As illustrated in Fig. 4, the sensitivities of both measures

¹⁵In line with this implication, [Brav et al. \(2018\)](#) find a negative relation between hedge fund activism and R&D expenditures in their sample of high-tech firms, while [Aghion et al. \(2013\)](#) report a strong positive relation between institutional ownership and R&D in a broad cross-section. In addition, [David et al. \(2001\)](#) find a positive relation between institutional investor activism and R&D. The role of incentive pay as a substitute for corporate governance is documented, for example in [Core et al. \(1999\)](#) or, more recently, [Fahlenbrach \(2009\)](#).

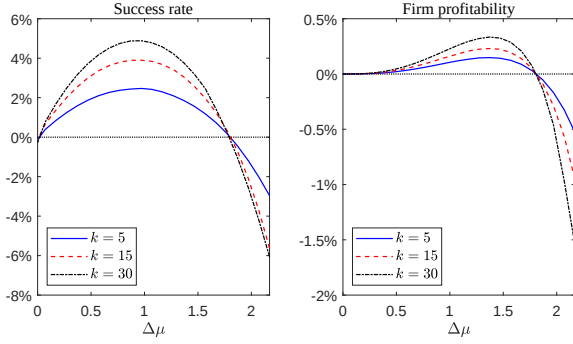


Fig. 4. Investment-cash sensitivities. We plot the average percentage change in success rate (left panel) and firm profitability (right panel) induced by a percentage change in liquidity (cash holdings), as a function of the returns to investment $\Delta\mu$. All parameters are set as in Example 1.

of investment output with respect to cash holdings increase in absolute terms with financing frictions. Importantly, however, the sign of this relation depends on the prevailing returns to investment.

Implication 4. *The sensitivity of investment output to cash flows is:*

- i) *positive and increasing in the costs of external financing in firms with low returns to investment, and*
- ii) *negative and decreasing in the costs of external financing in firms with high returns to investment.*

In light of Implication 4, the empirical results documenting the decline and disappearance of investment-cash flow sensitivities in the cross-sectional average over the last decades as discussed above, need not imply that investment-cash flow sensitivity is a bad measure of financial constraints (see, e.g., [Chen and Chen, 2012](#), for an argument along these lines). Instead, when investment input is not perfectly contractible, our model suggests that empirical tests of the dependence of intangible investment on cash flows in financially constrained firms should take into account that its sign may depend on the returns to intangible investment.

4.3. Implications for pay for investment performance

Given the considerable discretion managers have when deciding on how much to spend on investment in intangibles, executive compensation needs to provide incentives for them to invest in shareholders' best interest. In our model, the resulting agency costs create the distortions in and dynamics of intangible investment discussed above. However, our theoretical analysis provides clear predictions about how to optimally structure these investment incentives that ensure the firm's long-term profitability while also incentivizing the manager to meet short-term performance targets. In particular, we obtain novel empirical implications regarding the dependence of executive pay on the performance of intangible investment projects as captured

by the sensitivities β_t^s and β_t^b with respect to investment success and failure respectively.

Notably, in order to incentivize a given level of investment input at lowest costs, the optimal contract conditions executive pay mainly on the more informative investment outcome, resulting in an asymmetric response of compensation to investment success and failure. Since investment success is the less (more) informative performance signal when the implemented level of investment is high (low) (see Section 3.3) firms with high returns to investment stipulate an average reward for investment success, $\bar{\beta}^s$, that is smaller than the average punishment for investment failure, $|\bar{\beta}^b|$, while in firms with low returns to investment, this relation is reversed. This is illustrated in the upper right panel in Fig. 5.

Implication 5. *The average sensitivity of pay to investment output is asymmetric with pay being more sensitive to:*

- i) *investment success in firms with low returns to investment, and to*
- ii) *investment failure in firms with high returns to investment.*

To get some more intuition for Implication 5, note first that with higher returns to investment, the manager's compensation becomes more sensitive to investment output to incentivize higher investment expenditures (upper left panel of Fig. 5), i.e., the total sensitivity of pay to investment performance, $\bar{\beta}^s + |\bar{\beta}^b|$ increases in $\Delta\mu$ (see, e.g., [Balkin et al., 2000](#), for related empirical evidence). However, not only the size but also the structure of incentive compensation changes as investment becomes more profitable. In particular, as investment expenditures increase, investment failure becomes more and more informative. To provide investment incentives, the optimal contract, thus, stipulates on average stronger punishment following an investment failure (lower right panel of Fig. 5). In contrast, the average reward for investment success is hump-shaped in the returns to intangible investment (lower left panel of Fig. 5). That is, although the total sensitivity of compensation to investment output needs to increase in $\Delta\mu$ (upper left panel of Fig. 5), incentive provision via punishment for failure eventually becomes so efficient that the average reward for investment success decreases even in absolute terms and not only relative to the average punishment for failure.

The above implications on the sensitivity of pay with respect to intangible investment output focus on the part of the manager's incentive compensation that ensures the firm's long-term profitability. However, in our multitask setting, optimal incentive compensation needs to balance incentives for short- and long-term performance. More concretely, if a manager is harder to incentivize in his short-term task, which in our model is captured by higher benefits of cash flow diversion λ , reflecting for example weaker corporate governance, the sensitivity of pay with respect to short-term earnings has to increase. However, this also requires a higher sensitivity of pay with respect to the long-term investment outcome (see upper left panel Fig. 5), in order to avoid the manager from inflating short-term earnings by reducing long-term investment. Hence, under the optimal

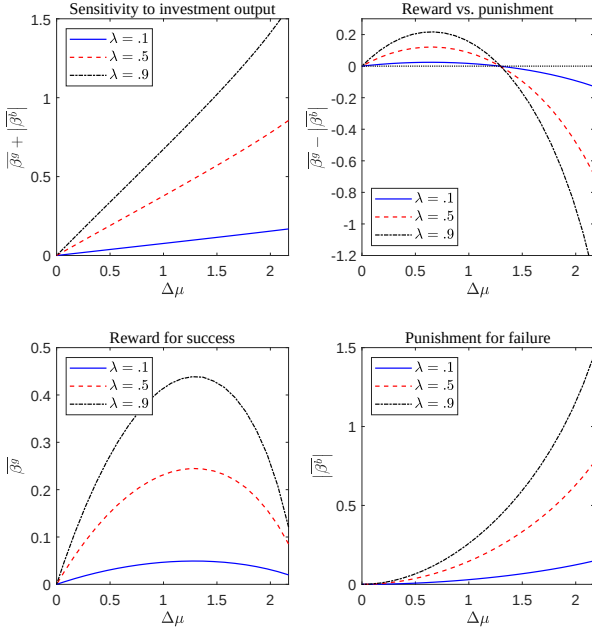


Fig. 5. Pay for investment performance. We plot the average reward and punishment as a function of the returns to investment $\Delta\mu$. All parameters are set as in Example 1.

contract, the sensitivity of pay to short-term performance and the sensitivity of pay to long-term performance are intimately linked.

Implication 6. *The average sensitivity of pay to investment output is increasing in the sensitivity of pay to short-term performance. This increase is asymmetric with pay becoming relatively more sensitive to:*

- i) *investment success in firms with low returns to investment, and to*
- ii) *investment failure in firms with high returns to investment.*

While a more severe cash flow diversion problem, thus, calls for a higher total sensitivity of pay to investment performance, the relative increase in rewards for investment success and punishments for investment failure again depends on the relative informativeness of these two investment outcomes. For firms with low returns to investment, incentivizing investment through rewards is more efficient than through punishment. Therefore, when an increase in diversion benefits λ (e.g., via a deterioration in corporate governance) requires steeper incentives also for investment, these firms increase the sensitivity of pay to investment success more than that with respect to investment failure. For firms with high returns to investment, the opposite applies. Overall, steeper short-term incentives, thus, call not only for steeper long-term incentives but also for a more asymmetric exposure to investment outcomes (upper right panel Fig. 5).

4.4. Implications for liquidity management

In the implementation of our model, financial frictions, which arise endogenously from the agency problem, provide a precautionary motive for corporate cash holdings or, more generally, liquidity management. In particular, fluctuations in corporate cash holdings via their dependence on both short-term cash flows as well as risky investment outcomes may require costly refinancing when cash reserves are depleted. One implication is that higher risk in a firm's cash position makes "financial slack" more valuable and, thus higher cash buffers are accumulated before the firm pays any dividends. Notably, the risk in a firm's cash position is endogenous in our multitask model. It varies with the returns to intangible investment via the implemented investment profile, as well as the pay for performance sensitivities to incentivize it. As a consequence, also corporate liquidity management strategies, such as payout policies, will vary systematically with the returns to intangible investment.

To see these relationships, consider the part of the instantaneous volatility of cash holdings, M , that stems from the exposure to investment output, $\sigma_M := (\beta^g - \beta^b) \sqrt{vp(1-p)}/\lambda$. In the left panel of Fig. 6, we plot the cross-sectional average of this endogenous volatility component under the optimal contract as a function of the returns to intangible investment. The endogenous volatility disappears for very low returns to investment, since i) only little exposure to the risky investment outcome is needed to incentivize investment expenditures close to zero (i.e., $\beta^g - \beta^b \rightarrow 0$ as $I \rightarrow 0$) and ii) the volatility of the investment outcome $p(1-p)$ vanishes as the probability of an investment success approaches zero. Now, as returns to investment increase, and with it optimal investment, also σ_M increases at first. Intuitively, the investment outcome becomes more risky (second term of σ_M increases) and the sensitivity of total pay to investment performance increases as well (first term of σ_M increases; see the upper left panel of Fig. 5). However, as returns to investment and with it the implemented investment level get sufficiently large, the volatility of the risky investment outcome, the second term of σ_M , decreases again since an investment failure becomes less and less likely. Eventually, this decrease will outweigh the increase in the pay for investment performance sensitivity, such that the average volatility of firms' cash holdings is hump-shaped in returns to investment (see left panel of Fig. 6).

Implication 7. *The average volatility of cash holdings is hump-shaped in returns to investment.*

Over the past decades, the cash holdings of US corporations have increased substantially, and this increase has gone hand in hand with a similar increase in the importance of intangible investment, in particular R&D (e.g., Pinkowitz et al., 2016). While this joint development has been attributed to the lower collateral value of intangible investments (see, e.g., Bates et al., 2009; Falato et al., 2021), our model highlights another, orthogonal channel, which operates via the agency problem between firm insiders and outside investors. The need to incentivize intangible investment requires exposure of the firm's cash position to additional risk, thus, creating additional endogenous

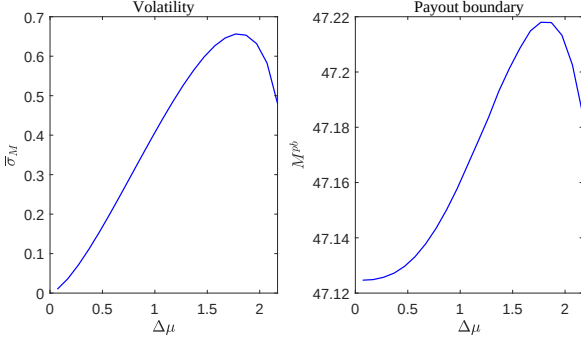


Fig. 6. Volatility and payout boundary. We plot the cross-sectional average instantaneous volatility in cash holdings due to providing investment incentives (left panel) and the payout boundary (right panel) as a function of the returns to investment $\Delta\mu$. All parameters are set as in Example 1.

volatility, which aggravates the precautionary motive for corporate cash holdings relative to an all-tangible investment firm. As a consequence, optimal corporate payout policies require a higher cash balance before distributing dividends to shareholders. In light of Implication 7, however, the relationship between the importance of intangible investment and precautionary cash holdings is not monotonic but follows the same shape as the endogenous volatility component σ_M . Hence, as returns to investment become sufficiently high, the target level of cash holdings at which dividends are distributed eventually decreases again. This pattern is illustrated in the right panel of Fig. 6, which shows the optimal payout barrier M^{pb} (i.e., the level of cash holdings at which firms start paying dividends).

Implication 8. *The average cash holdings of dividend paying firms are hump-shaped in returns to investment.*

5. Conclusion

In this paper, we analyze the dynamics of intangible (non-contractible) investment under endogenous financial constraints in a dynamic agency model with multitasking. The manager privately observes the firm's cash flows, which he can divert for private consumption and, in addition, use to finance non-contractible investment in the firm's future profitability. Thus, when reported cash flows are low (or even negative), the principal does not know whether this is due to stealing or investment. To ensure truth-telling and investment in the interest of firm owners, the optimal compensation scheme ties the manager's compensation both to reported cash flows and imperfect performance signals indicating investment success or failure. Exposing the manager to compensation risk is, however, costly, such that incentivizing investment causes additional agency costs.

The optimal investment profile trades off these agency costs of investment with the potential efficiency gains. As a result, investment will be history dependent and distorted away from its first-best level. Investment distortions decrease (in absolute terms) with good performance, as the manager's stake in the

firm increases mitigating the agency problem. Our model predicts overinvestment and negative investment-cash flow sensitivities in industries where returns to non-contractible intangible investment such as R&D are high (e.g., as being in a position of technological leadership is important like in the pharmaceutical industry). By contrast, in industries where this is not the case, investment will be distorted downwards and increasing in cash flows.

If raising external funds is more expensive, investment will be distorted more severely and more sensitive to past performance; in firms with high returns to investment, investment increases and the relation between financial slack and investment becomes more negative, while investment decreases and its sensitivity to financial slack becomes more positive in firms with low returns to investment. Our multitask theory, thus, offers a new perspective on the interpretation of investment-cash flow sensitivities as a measure of financial constraints, taking into account the increased importance of alternative forms of investment such as process innovation, R&D, or intangible investment in general. Furthermore, we provide new, testable implications on how the structure of executive compensation, as well as level and volatility of corporate cash holdings differ between industries in which intangible investment is of high, respectively low importance.

Appendix A. Omitted proofs

A.1. Proof of Lemma 1

For $S = S^*$, we define the manager's expected lifetime utility evaluated conditional on time t information by:

$$v_t = \int_0^t e^{-\gamma s} dU_s + e^{-\gamma t} w_t, \quad (\text{A.1})$$

which, by construction, is a martingale with respect to the filtration generated by (Z, N^j) , $j \in \{g, b\}$ under the probability measure Q^{S^*} . By the martingale representation theorem (see, Theorem III 4.34 in Jacod and Shiryaev, 2003), we can express v_t for some predictable processes α and β^j , $j \in \{g, b\}$ as:

$$\begin{aligned} v_t = v_0 &+ \int_0^t e^{-\gamma s} \alpha_s (d\hat{Y}_s - (\mu_s - I_s^*) ds) \\ &+ \int_0^t e^{-\gamma s} \beta_s^g (dN_s^g - \nu p(I_s^*) ds) \\ &+ \int_0^t e^{-\gamma s} \beta_s^b (dN_s^b - \nu(1 - p(I_s^*)) ds), \end{aligned}$$

where, under Q^{S^*} , $\hat{Y}_t - \int_0^t (\mu_s - I_s^*) ds$ is a standard Brownian motion and $N_t^g - \int_0^t \nu p(I_s^*) ds$, and $N_t^b - \int_0^t \nu(1 - p(I_s^*)) ds$ are compensated Poisson processes. Differentiating both this representation and the definition of v_t in (A.1) yields (6). $\square$

A.2. Proof of Lemma 2

Consider any feasible policy of the agent, $S = \{\hat{Y}_t, I_t : 0 \leq t \leq \tau\}$, with $dY_t - d\hat{Y}_t \geq 0$ and $I_t \geq 0$. The asso-

ciated expected lifetime utility is given by:

$$\begin{aligned} v_t = v_0 &+ \int_0^t e^{-\gamma s} \alpha_s (d\hat{Y}_s - (\mu_s - I_s^*) ds) \\ &+ \int_0^t e^{-\gamma s} \lambda (dY_s - d\hat{Y}_s) \\ &+ \int_0^t e^{-\gamma s} \beta_s^g (dN_s^g - \nu p(I_s^*) ds) \\ &+ \int_0^t e^{-\gamma s} \beta_s^b (dN_s^b - \nu (1 - p(I_s^*)) ds), \end{aligned} \quad (\text{A.2})$$

where $d\hat{Y}_t - (\mu_t - I_t^*) dt = \sigma dZ_t$ for $S = S^*$, and dN_t^g (dN_t^b) is a Poisson process with intensity $\nu p(I_t)$ ($\nu (1 - p(I_t))$). Differentiating (A.2) and taking expectations gives:

$$\begin{aligned} e^{\gamma t} E[dv_t] &= (\lambda - \alpha_t) E[dY_t - d\hat{Y}_t] + \alpha_t (I_t^* - I_t) dt \\ &+ \nu (\beta_t^g - \beta_t^b) (p(I_t) - p(I_t^*)) dt. \end{aligned}$$

Clearly, v_t is a martingale for $S = S^*$. For $S = S^*$ to be incentive compatible, the drift of v_t has to be non-positive for all possible deviations (i.e., v_t has to be a supermartingale for any feasible $S \neq S^*$). First consider a deviation from truth-telling (i.e., underreporting for consumption $dY_t - d\hat{Y}_t > 0$). This deviation is suboptimal for the agent if $\alpha_t \geq \lambda$. Second, increasing I_t marginally above I_t^* , is suboptimal for the agent if $\alpha_t \geq \nu p'(I_t) (\beta_t^g - \beta_t^b)$, while, third, decreasing I_t marginally below I_t^* is suboptimal if $\alpha_t \leq \nu p'(I_t) (\beta_t^g - \beta_t^b)$. So, incentive compatibility requires (7) and (8) to hold. $\square$

A.3. Proof of Lemma 3

Consider the process κ counting the number of replacements and the associated time points $\tau(\kappa)$ where we normalize $\tau(0) = 0$. Then we can write the principal's value as:

$$f^i(w) = \max_{\mathbf{I}, \mathbf{U}, \tau} \left\{ E \left[\int_0^\infty e^{-rs} (dY_s - dU_s) - k \sum_{\kappa=1}^\infty e^{-r\tau(\kappa)} \middle| w, \mu^i \right] \right\},$$

where maximization is subject to incentive compatibility in (4), limited liability, as well as promise-keeping, $w_0 = w$. Now denote the (stochastic) time of the next technology shock by T and recall that technology shocks arrive with exogenous rate, such that T is clearly beyond both the agent's and the principal's control. Thus, substituting from (1), we have:

$$\begin{aligned} f^i(w) &= \frac{\mu^i}{r + \nu} + \max_{\mathbf{I}, \mathbf{U}, \tau} \left\{ E \left[\int_T^\infty e^{-rs} \mu_s ds \right. \right. \\ &\quad \left. \left. - \int_0^\infty e^{-rs} (dU_s + I_s ds) - k \sum_{\kappa=1}^\infty e^{-r\tau(\kappa)} \middle| w \right] \right\}, \end{aligned}$$

where the maximization problem determining the optimal contract is clearly independent of $\mu_0 = \mu^i$ for any w . $\square$

A.4. Proof of Proposition 1

We first show that the function $f(w)$ is concave in w and then verify that the contract of Proposition 1 maximizes the principal's expected payoff.

A.4.1. Concavity

Consider the function $f(w)$ that, for a given w and μ , solves

$$\begin{aligned} (r + \nu) f(w) &= \mu + \nu p(I(w)) \Delta - I(w) \\ &+ \frac{1}{2} (\alpha(w))^2 \sigma^2 f''(w) \\ &+ [\gamma w - \nu (\beta^g(w) p(I(w)) \\ &+ \beta^b(w) (1 - p(I(w))))] f'(w) \\ &+ \nu p(I(w)) f(w + \beta^g(w)) \\ &+ \nu (1 - p(I(w))) f(w + \beta^b(w)) \end{aligned} \quad (\text{A.3})$$

for $w \in [0, \bar{w}]$ and $f(w) = f(\bar{w}) - (w - \bar{w})$ for $w > \bar{w}$, with boundary conditions $f(0) = \max_w \{f(w) - k\}$, $f'(\bar{w}) = -1$, and $f''(\bar{w}) = 0$.

Investment $I(w)$ satisfies the following first-order condition:

$$\begin{aligned} 1 &= \nu p'(I(w)) (\beta^g(w) - \beta^b(w)) f_w'(w) \\ &+ \nu p'(I(w)) [f(w + \beta^g(w)) + \Delta - f(w + \beta^b(w))] \\ &+ \alpha(w) p(I(w)) \frac{p''(I(w))}{p'(I(w))^2} [f'(w) - f'(w + \beta^g(w))], \end{aligned} \quad (\text{A.4})$$

and $\beta^i(w)$, $i \in \{g, b\}$ is determined from (8) together with:

$$\begin{aligned} f'(w) &= p(I(w)) f'(w + \beta^g(w)) \\ &+ (1 - p(I(w))) f'(w + \beta^b(w)) \end{aligned} \quad (\text{A.5})$$

when $\beta^b(w) > -w$ and together with $\beta^b(w) = -w$ otherwise.

Differentiating (A.3) and using (A.4) and (A.5) yields:

$$\begin{aligned} -(\gamma - r) f'(w) &= \left\{ \gamma w - \nu \left[\frac{\alpha(w) p(I(w))}{\nu p'(I(w))} + \beta^b(w) \right] \right\} f''(w) \\ &+ \frac{1}{2} \sigma^2 (\alpha(w))^2 f'''(w). \end{aligned}$$

From the boundary conditions at $\bar{w}$, we get $f'''(\bar{w}) = 2 \frac{\gamma - r}{\sigma^2 \alpha(\bar{w})^2} > 0$, such that $\exists \varepsilon > 0$ with $f'''(\bar{w} - \varepsilon) > 0$ and $f''(\bar{w} - \varepsilon) < 0$.

The proof then is by contradiction. So assume that $\exists \check{w} := \sup\{w < \bar{w} : f''(w) \geq 0\}$, where it holds by continuity that $f''(\check{w}) = 0$ and $f'''(\check{w}) < 0$, implying that $f'(\check{w}) = -f'''(\check{w}) \frac{1}{2} \frac{\sigma^2 \alpha(\check{w})^2}{\gamma - r} > 0$. Now, consider two points $w^1 < \check{w} < w^2$ close to $\check{w}$, such that $f''(w^1) > 0 > f''(w^2)$ and $w^1 f'(w^1) = w^2 f'(w^2)$ and observe that $f(w)$ can be written as:

$$rf(w) = \gamma w f'(w) + \frac{1}{2} \sigma^2 (\alpha(w))^2 f''(w) + g(w),$$

with

$$\begin{aligned} g(w) &= \mu - I(w) \\ &+ \nu \left\{ p(I(w)) \left[f(w + \beta^b(w) + \frac{\alpha(w)}{\nu p'(I(w))}) + \Delta \right] \right. \\ &+ (1 - p(I(w))) f(w + \beta^b(w)) - f(w) \\ &\left. - \left[\frac{\alpha(w) p(I(w))}{\nu p'(I(w))} + \beta^b(w) \right] f'(w) \right\}. \end{aligned}$$

Next, compute the differential of $g(w)$ around $\check{w}$, $dg(w)|_{w=\check{w}} = g'(\check{w})dw$, and observe that

$$\begin{aligned} g'(\check{w}) &= \partial I(w, \mu) / \partial w \left\{ -1 - \alpha(w) f'(\check{w}) \right. \\ &\quad + \nu p'(I(w)) [f(\check{w} + \beta^g(\check{w})) + \Delta - f(\check{w} + \beta^b(w))] \\ &\quad + \frac{\alpha(w) p(I(w)) p''(I(w))}{p'(I(w))^2} [f'(\check{w}) - f'(\check{w} + \beta^g(w))] \left. \right\} \\ &\quad + \nu (1 + \partial \beta^b(w) / \partial w) \left\{ p(I(w)) f'(\check{w} + \beta^g(w)) \right. \\ &\quad + (1 - p(I(w))) f'(\check{w} + \beta^b(w)) - f'(\check{w}) \left. \right\} \\ &\quad - \nu \left[\frac{\alpha(w) p(I(w))}{\nu p'(I(w))} + \beta^b(w) \right] f''(\check{w}) \\ &= 0, \end{aligned}$$

which follows from $f''(\check{w}) = 0$, together with (A.4) and either (A.5), or, in case the limited liability constraint binds, $\partial \beta^b(w) / \partial w = -1$. Thus, evaluating $f(w)$ in w^1 and w^2 , we get:

$$r[f(w^1) - f(w^2)] = \frac{1}{2} \sigma^2 \alpha(w)^2 [f''(w^1) - f''(w^2)] > 0,$$

where we have used that the effect of a change in $g(w)$ around $\check{w}$ is of second order and will thus be dominated by the change in $f''(w)$. However, this directly contradicts $f'(\check{w}) > 0$. $\square$

A.4.2. Verification

For any incentive compatible contract (S, U, τ) , define:

$$G_t = \int_0^t e^{-rs} (dY_s - dU_s) + e^{-rt} f(w_{t-}, \mu_t)$$

and recall that w_{t-} evolves according to:

$$\begin{aligned} dw_{t-} &= \gamma w_{t-} dt - dU_t + \alpha_t \sigma dZ_t \\ &\quad + \beta_t^g (dN_t^g - \nu p(I_t) dt) \\ &\quad + \beta_t^b (dN_t^b - \nu (1 - p(I_t)) dt), \end{aligned}$$

where $\alpha_t \geq \lambda$ and $\beta_t^g - \beta_t^b = \frac{\alpha_t}{\nu p'(I_t)}$. Recall that $f(w) = f(w, \mu^l) = f(w, \mu^h) - \Delta$, so that from differentiating G using Itô's lemma and the change in variables formula for point pro-

cesses as well as (A.3), we get:

$$\begin{aligned} e^{rt} dG_t &= \left\{ \nu \left[p(I_t) [f(w_{t-} + \beta_t^g) + \Delta] + (1 - p(I_t)) f(w_{t-} + \beta_t^b) \right. \right. \\ &\quad - \left[\beta_t^g p(I_t) + \beta_t^b (1 - p(I_t)) \right] f'(w_{t-}) - I_t \left. \right] dt \\ &\quad - \nu \left[p(I(w)) [f(w + \beta^g(w)) + \Delta] \right. \\ &\quad + (1 - p(I(w))) f(w + \beta^b(w)) - I(w) \\ &\quad - \left[\beta^g(w) p(I(w)) + \beta^b(w) (1 - p(I(w))) \right] f'(w) \left. \right] \left. \right\} dt \\ &\quad + \frac{1}{2} (\alpha_t^2 - \lambda^2) \sigma^2 f''(w_{t-}) dt - [1 + f'(w_{t-})] dU_t \\ &\quad + [\sigma + \alpha_t \sigma f'(w_{t-})] dZ_t \\ &\quad + [f(w_{t-} + \beta_t^g) + \Delta] d\hat{N}_t^g + f(w_{t-} + \beta_t^b) d\hat{N}_t^b \\ &\quad - f(w_{t-}) (d\hat{N}_t^g + d\hat{N}_t^b) \end{aligned} \tag{A.6}$$

for $w = w_{t-} \in [0, \bar{w}]$, where $\hat{N}^i$ denotes the compensated point processes associated with N^i , $i \in \{g, b\}$. Now observe that the term in curly brackets is less than or equal to zero, because $\beta^i(w)$, $i \in \{g, b\}$, and $I(w)$ are the solution to:

$$\begin{aligned} \max_{\beta^g \geq -w, I} & \left[p(I) [f(w + \beta^g) + \Delta] + (1 - p(I)) f(w + \beta^b) \right. \\ & \left. - (\beta^g p(I) + \beta^b (1 - p(I))) f'(w) - I \right]. \end{aligned}$$

Now we turn to the remaining terms in (A.6) chronologically. The first term is non positive as $f'' \leq 0$ and $\alpha_t \geq \lambda$ for any $t \geq 0$. The second term is non positive as $f' \geq -1$ and $dU \geq 0$ while Z and $\hat{N}^i$ are martingales. Hence, G_t is a supermartingale and a martingale if and only if for $t > 0$, $\beta_t^i = \beta^i(w)$, $I_t = I(w)$, $\alpha_t = \lambda$, and $dU_t > 0$ only when $w > \bar{w}$. Now consider the principal's expected payoff under any incentive compatible contract (S, U, τ) :

$$E \left[\int_0^\tau e^{-rs} (dY_s - dU_s) + e^{r\tau} L_\tau \right],$$

where τ denotes the time when the incumbent manager is replaced and L_τ the principal's expected profits from restarting with a new agent, net of replacement costs k . We then have

that:

$$\begin{aligned}
& E \left[\int_0^\tau e^{-rt} (dY_s - dU_s) + e^{r\tau} L_\tau, \right] \\
&= E [G_{t \wedge \tau}] \\
&+ E \left[\mathbf{1}_{t \leq \tau} \left(\int_t^\tau e^{-rs} (dY_s - dU_s) + e^{-r\tau} L_\tau - e^{-rt} f(w_{t-}) \right) \right] \\
&\leq f(w_{0-}) \\
&+ E \left[\mathbf{1}_{t \leq \tau} \left(\int_t^\tau e^{-rs} (dY_s - dU_s) + e^{-r\tau} L_\tau - e^{-rt} f(w_{t-}) \right) \right] \\
&= f(w_{0-}) \\
&+ e^{-rt} E \left[\mathbf{1}_{t \leq \tau} \left\{ E \left[\int_t^\tau e^{-r(s-t)} (dY_s - dU_s) + e^{-r(\tau-t)} L_\tau \middle| \mathcal{F}_t \right] \right. \right. \\
&\quad \left. \left. - f(w_{t-}) \right\} \right],
\end{aligned}$$

where the inequality follows since $G_{t \wedge \tau}$ is a supermartingale and $G_0 = f(w_{0-})$. Now note that:

$$E \left[\int_t^\tau e^{-r(s-t)} (dY_s - dU_s) + e^{-r(\tau-t)} L_\tau \middle| \mathcal{F}_t \right] < \frac{\mu^h}{r} - w_{t-},$$

and, as $f' \geq -1$, we have that $f(w_{t-}) + w_{t-} \geq L_\tau$, which yields:

$$\begin{aligned}
E \left[\int_0^\tau e^{-rt} (dY_s - dU_s) + e^{r\tau} L_\tau, \right] &\leq f(w_{0-}) \\
&+ e^{-rt} E \left[\mathbf{1}_{t \leq \tau} \left(\frac{\mu^h}{r} - L_\tau \right) \right].
\end{aligned}$$

Taking $t \rightarrow \infty$ yields:

$$E \left[\int_0^\tau e^{-rt} (dY_s - dU_s) + e^{r\tau} L_\tau, \right] \leq f(w_{0-}). \quad (\text{A.7})$$

Finally, under the contract stated in Proposition 1, $G_{t \wedge \tau}$ is a martingale and, hence, (A.7) holds with equality. $\square$

A.5. Proof of Lemma 4

Whenever the limited liability constraint (9) does not bind, for which it is sufficient that $w \geq \frac{\lambda}{vp'(I)}$, first-order condition (19) holds. The first claim then follows immediately from rewriting (19) as:

$$\frac{f'(w + \beta^b(w)) - f'(w)}{f'(w) - f'(w + \beta^g(w))} = \frac{p(I(w))}{1 - p(I(w))}, \quad (\text{A.8})$$

and observing that $p(I(w))|_{\psi'} < p(I(w))|_{\psi''}$ since $I(w)|_{\psi'} < I(w)|_{\psi''}$ and $p'(\cdot) > 0$. Next, when $I(w)|_{\psi} \rightarrow \bar{I}$ and the limited liability constraint is slack, the strict concavity of $f(w)$ implies that (19) can only be satisfied for $\beta^g(w) \rightarrow 0$. In case $p(I) \rightarrow 0$, (19) similarly requires that $\beta^b(w) \rightarrow 0$, while, from incentive compatibility, $\beta^g(w) \rightarrow \frac{\lambda}{vp'(0)}$. For bounded $p'(0)$, it then trivially follows that $-\beta^b(w)/\beta^g(w)$ goes to zero as $I(w) \rightarrow 0$. Now, if $p'(I) \rightarrow \infty$ as $I \rightarrow 0$, both $\beta^g(w)$ and $\beta^b(w)$ go to zero as $I \rightarrow 0$. Then, up to a first-order approximation, we have $f'(w + \beta^b(w)) = f'(w) + \beta^b(w)f''(w)$, and

$f'(w + \beta^g(w)) = f'(w) + \beta^g(w)f''(w)$. Hence, there exists an $\varepsilon > 0$ such that, from (A.8), we have $\frac{p(\varepsilon)}{1-p(\varepsilon)} = \frac{-\beta^b(w)}{\beta^g(w)}$.

Finally, the last claim follows from strict concavity of $f(w)$ for $w < \bar{w}$, implying that (19) can only be satisfied for $\beta^g > 0 > \beta^b$. $\square$

A.6. Proof of Lemma 5

See main text. $\square$

A.7. Proof of Proposition 2

Denote the right-hand side of the HJB in (12) by $\mathcal{L}$ and take the first derivative with respect to I , noting that $\beta^g = \beta^b + \frac{\lambda}{vp'(I)}$ from incentive constraint (8), to obtain:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial I} &= vp'(I(w))\Delta - 1 - \left[\left(\frac{\lambda}{v} \frac{-p''(I(w))}{[p'(I(w))]^2} \right) \delta^{gb}(w) \right. \\
&\quad \left. + vp'(I(w))(\delta^g(w) - \delta^b(w)) \right],
\end{aligned} \quad (\text{A.9})$$

with

$$\begin{aligned}
\delta^{gb}(w) &:= p(I(w))(1 - p(I(w))) \\
&\quad \cdot v \left[f'(w + \beta^b(w)) - f'(w + \beta^g(w)) \right] \\
&\geq 0,
\end{aligned} \quad (\text{A.10})$$

$$\begin{aligned}
\delta^j(w) &:= f(w) + \beta^j(w)f'(w) - f(w + \beta^j(w)) \\
&\geq 0, \quad j \in \{g, b\},
\end{aligned} \quad (\text{A.11})$$

where we have used (19), which holds whenever the limited liability constraint does not bind, and the inequalities in (A.10) and (A.11) follow from strict concavity of $f(w)$ for $w < \bar{w}$. The term in square brackets in (A.9) are the marginal agency costs of investment, which we denote by $\phi(w)$.

To show the overinvestment result for sufficiently high levels of I_{FB} , it is sufficient to show that the marginal agency costs of investment are strictly negative as I_{FB} approaches $\bar{I}$, such that $\partial \mathcal{L} / \partial I|_{I(w)=I_{FB}} > 0$ for $I_{FB} \nearrow \bar{I}$. The result then follows from continuity and the fact that $I(w) = I_{FB}$ is the uniquely optimal investment level for $I_{FB} = \bar{I}$. So, note that $p(\bar{I}) = 1$ implies that $\delta^{gb}(w) \rightarrow 0$ and, by Lemma 5, $\beta^g(w), \delta^g(w) \rightarrow 0$, such that $\phi(w)|_{I(w)=I_{FB}} \rightarrow vp'(I_{FB})\delta^b(w) < 0$, where the inequality follows again from the concavity of $f(w)$ and the fact that $\beta^b(w) > 0$ by incentive constraint (8). Note also that from $vp'(I_{FB})\Delta - 1 = 0$, we must have that $p'(I_{FB}) > 0$, which is also a necessary condition for limited liability not to bind.

As for the remaining case, underinvestment if I_{FB} is sufficiently low, we show that $\partial \mathcal{L} / \partial I|_{I(w)=I_{FB}} \rightarrow 0$ from below as $I_{FB} \searrow 0$. To see this note that, in the limit, δ^g dominates δ^b from Lemma 5, such that $\phi(w)$ is strictly positive. The result then follows from continuity and the fact that $I(w) = I_{FB}$ is the uniquely optimal investment level for $I_{FB} = 0$. $\square$

A.8. Proof of Lemma 6

For part i), we first show that there exists $\widehat{w}_B \in (0, \bar{w})$, such that $\beta^b(w) = -w$ for $w \in [0, \widehat{w}_B]$. To see this, note that, for any $I(w) > 0$, we must have $\beta^g(w) - \beta^b(w) > 0$, which together with

the strict concavity of $f(w)$ for $w \in [0, \bar{w}]$ implies that the first-order condition in (19) cannot be satisfied in a neighborhood of $w = 0$. Thus, we get the stated result, with $\beta^b(w) = -w$ and $\beta^g(w) = \beta^b + \frac{\lambda}{vp'(I(w))} > 0$ on $w \in [0, \bar{w}_\beta]$, where:

$$\bar{w}_\beta := \min \left\{ w > 0 : p(I(w))f' \left(\frac{\lambda}{vp'(I(w))} \right) + (1 - p(I(w)))f'(0) = f'(w) \right\}.$$

Next, let us show the underinvestment result for small w . With the limited liability constraint binding, marginal agency costs of investment $\phi(w)$ as defined in the proof of Proposition 2 can be rewritten as follows:

$$\begin{aligned} \phi(w) = & vp'(I(w)) \left[f \left(\frac{\lambda}{vp'(I(w))} \right) - \left(f(0) + \frac{\lambda}{vp'(I(w))} f'(w) \right) \right] \\ & + \lambda \frac{p(I(w))p''(I(w))}{(p'(I(w)))^2} [f'(w) - f'(w + \beta^g(w))]. \end{aligned}$$

It remains to show that this expression is negative for small w . To see this, note first that the expression in the second line is negative for all w by strict concavity of $f(w)$. As for the remaining right-hand-side expression, it also follows from concavity of $f(w)$ that this is negative for w small enough.

For part iii), note that as $f(w)$ extends linearly for $w > \bar{w}$, we have that $f'(\bar{w} + \beta^g(\bar{w})) = f'(\bar{w})$ and (19) can only be satisfied if $\beta^b(\bar{w}) = 0$. It remains to show that $I(\bar{w}) = I_{FB}$. Since $\beta^g(\bar{w}) > 0$, but $f'(\bar{w} + \beta^g(\bar{w})) = f'(\bar{w})$, and $\beta^b(\bar{w}) = 0$, the marginal agency costs of investment $\phi(w)$ as defined in the proof of Proposition 2 are equal to zero, implying first-best investment.

Part ii) then is immediate as $\beta^b(\bar{w}) = 0$ and continuity implies that the limited liability constraint (9) is slack for w close to $\bar{w}$, such that Lemma 4 and Proposition 2 apply. $\square$

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